

The simplest systems containing a second-degree equation

One equation is linear — so make one unknown the subject and substitute.

LESSON 28–29

2 × 40 MIN

ALGEBRA 9, §13

IGX 10.2

LESSON CLOCK

11'

22'

20'

20'

7'

What a solution is	11 min
Substitution	22 min
How many solutions	20 min
Reading the picture	20 min
Homework	7 min

LEARNING OBJECTIVES

- Solve a linear-and-quadratic system by substitution.
- Interpret the solutions as the meeting points of a line and a curve.
- Use the discriminant to say how many solutions there are.
- Give every solution as an ordered pair.

TEXTBOOK

National	pp. 68–71
Cambridge	Core & Extended, pp. 216–225
Workbook	Exercise 10.2

01

Explanation

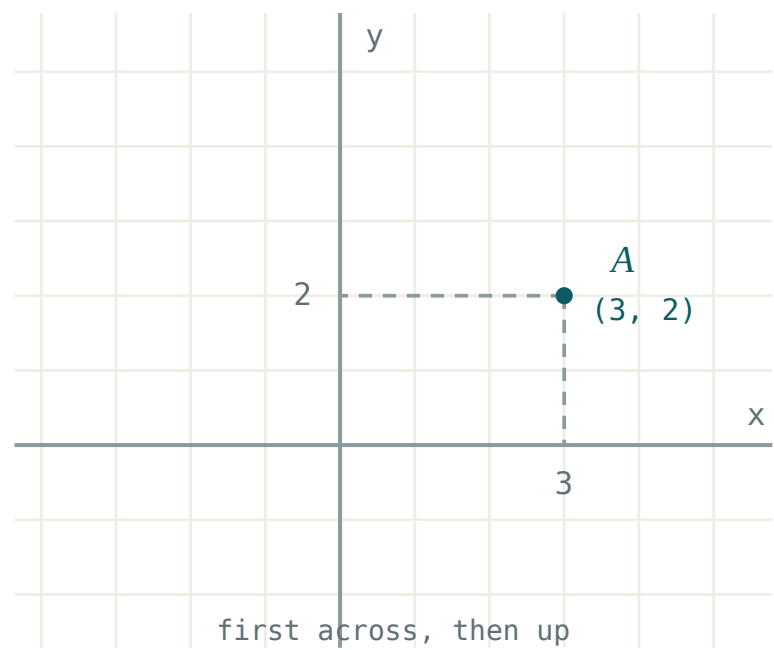
What a solution is

A solution of a system in two unknowns is an **ordered pair** (x, y) that satisfies **both** equations at once.

$\{ y = x + 1 ; x^2 + y^2 = 25 \}$ has solutions $(3, 4)$ and $(-4, -3)$

TWO NUMBERS, PAIRED – NOT TWO SEPARATE ANSWERS

Writing “ $x = 3, -4$ and $y = 4, -3$ ” is ambiguous: it seems to allow $(3, -3)$, which satisfies neither equation. Always write the pairs.



A solution is one point of the plane, lying on both graphs at once.

Substitution

THE METHOD, WHEN ONE EQUATION IS LINEAR

1. From the **linear** equation, make the easier unknown the subject.

2. Substitute into the second-degree equation.

3. Solve the resulting quadratic in one unknown.

4. Substitute each root back into the **linear** equation for the partner.

Step 4 uses the linear equation, never the quadratic — it is simpler and cannot introduce a false pair.

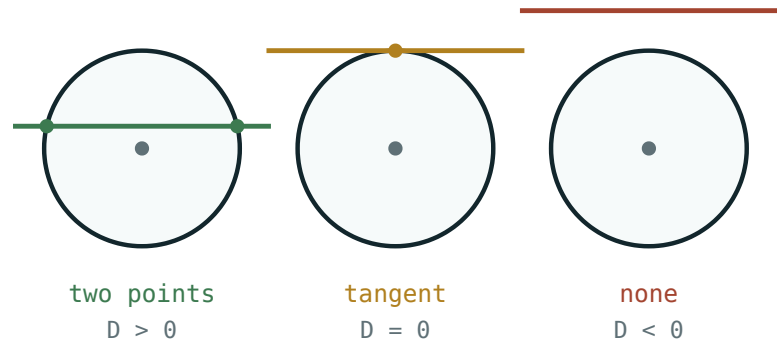
Example. { $x + y = 5$; $xy = 6$ }.

STEP	WORKING
make the subject	$y = 5 - x$
substitute	$x(5 - x) = 6$
tidy	$x^2 - 5x + 6 = 0$
solve	$x = 2, 3$
partners	$(2, 3)$ and $(3, 2)$

How many solutions

After substitution the system becomes one quadratic, so the discriminant answers the counting question directly.

D	SOLUTIONS	GEOMETRICALLY
$D > 0$	two pairs	the line cuts the curve twice
$D = 0$	one pair	the line is a tangent
$D < 0$	none	they miss each other



Two points, one, or none — decided by D alone.

Example. For which c is $y = x + c$ a tangent to $x^2 + y^2 = 8$? Substituting: $2x^2 + 2cx + c^2 - 8 = 0$, so $D = 4c^2 - 8(c^2 - 8) = 64 - 4c^2$. Setting $D = 0$ gives $c = \pm 4$.

Reading the picture

SYSTEM	THE TWO GRAPHS	SOLUTIONS
$y = x + 1 ; x^2 + y^2 = 25$	line and circle	$(3, 4), (-4, -3)$
$y = 2x ; y = x^2$	line and parabola	$(0, 0), (2, 4)$
$x + y = 5 ; xy = 6$	line and hyperbola	$(2, 3), (3, 2)$
$y = x + 5 ; y = x^2$	line and parabola	two irrational pairs

ALWAYS CHECK ONE PAIR

Substituting $(3, 4)$ into both equations of the first system takes ten seconds and catches every arithmetic slip. The check is part of the answer, not an optional extra.

02 Worked examples

EXAMPLE 1 Solve $\{x + y = 5 ; xy = 6\}$.

① $y = 5 - x$

From the linear equation.

② $x(5 - x) = 6 \Rightarrow x^2 - 5x + 6 = 0$

③ $x = 2$ or $x = 3$

④ $(2, 3)$ and $(3, 2)$.

Check: $2 + 3 = 5$, $2 \times 3 = 6$ ✓

ANSWER

$(2, 3)$ and $(3, 2)$

EXAMPLE 2 Solve $\{y = x + 1 ; x^2 + y^2 = 25\}$.

① $x^2 + (x + 1)^2 = 25$

② $2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0$

③ $x = 3$ or $x = -4$

④ $(3, 4)$ and $(-4, -3)$.

Partners from the line.

ANSWER

$(3, 4)$ and $(-4, -3)$

EXAMPLE 3 For which c is $y = x + c$ a tangent to $x^2 + y^2 = 8$?

① $x^2 + (x + c)^2 = 8$

② $2x^2 + 2cx + c^2 - 8 = 0$

③ $D = 4c^2 - 8(c^2 - 8) = 64 - 4c^2$

④ $D = 0 \Rightarrow c = \pm 4$

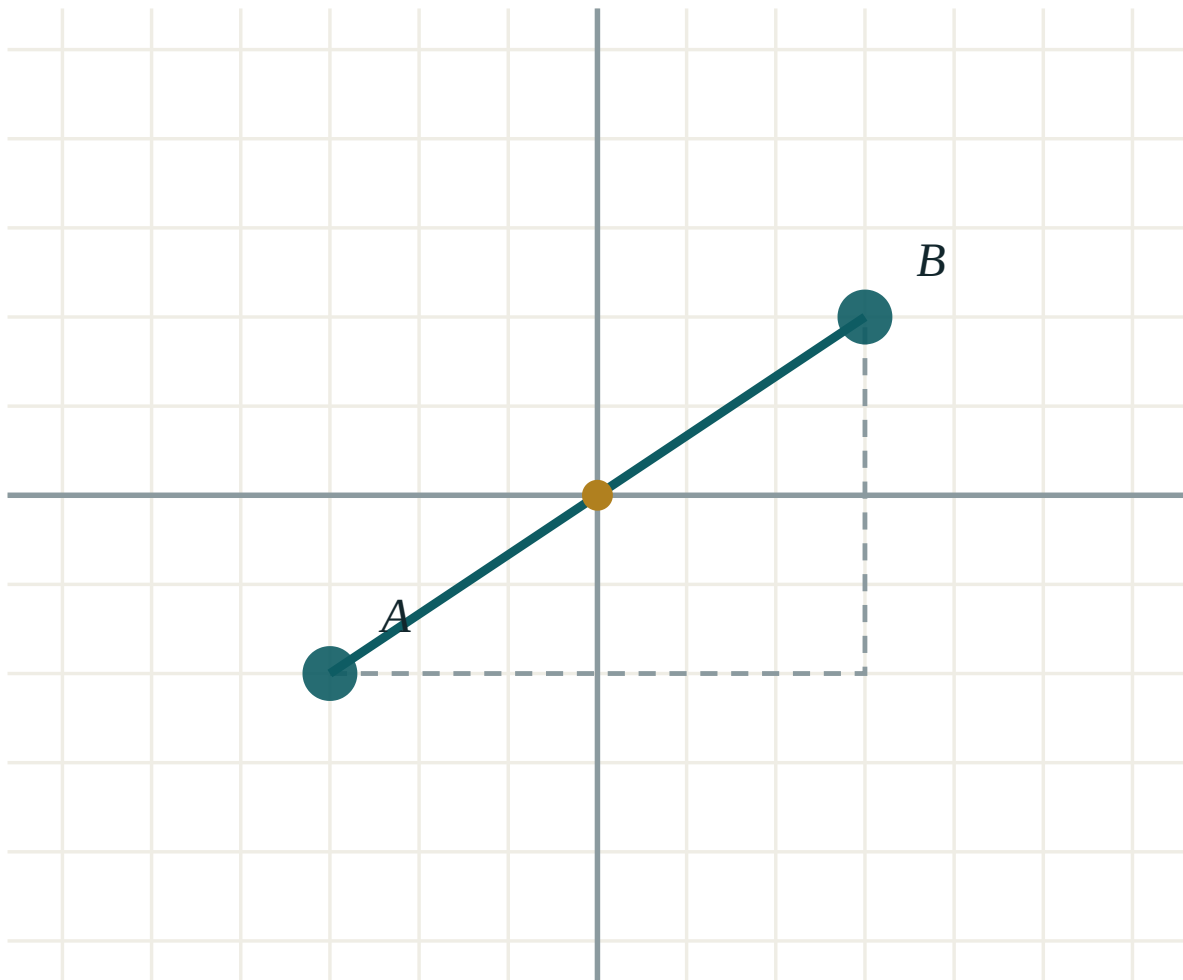
ANSWER

$c = 4$ or $c = -4$

03 Interactive model

Draw the line and the circle on the board and move the line; the class calls out two, one or none before any algebra.

Where two graphs meet



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
System of equations	Tenglamalar sistemasi	Система уравнений
Substitution method	O'rniga qo'yish usuli	Способ подстановки
Second-degree equation	Ikkinchi darajali tenglama	Уравнение второй степени
Ordered pair	Tartiblangan juftlik	Упорядоченная пара
Point of intersection	Kesishish nuqtasi	Точка пересечения
Consistent system	Birgalikdagi sistema	Совместная система
No solution	Yechimga ega emas	Не имеет решений
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A solution of a system in two unknowns is:

a number

an ordered pair

two separate numbers

an interval

With one linear equation, use:

elimination

substitution

squaring

a graph only

Substitute the roots back into:

the quadratic

the linear equation

either

neither

$D = 0$ after substitution means:

two points

a tangent

no meeting

an error

$\{ x + y = 5 ; xy = 6 \}$ has:

one solution

two solutions

none

infinitely many

The check is:

optional

part of the answer

only for hard questions

impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Make y the subject of $x + y = 5$

ANSWER $y = 5 - x$

2. Solve $\{ y = x ; y = x^2 \}$

ANSWER $(0, 0), (1, 1)$

3. Solve $\{ y = 2x ; y = x^2 \}$

ANSWER $(0, 0), (2, 4)$

4. Solve $\{ y = 4 ; y = x^2 \}$

ANSWER $(\pm 2, 4)$

5. Does $(3, 4)$ satisfy $x^2 + y^2 = 25$?

ANSWER Yes

6. Does $(3, 4)$ satisfy $y = x + 1$?

ANSWER Yes

7. Solve $\{ x = 2 ; x^2 + y^2 = 13 \}$

ANSWER $(2, \pm 3)$

Medium · the standard question

7 PROBLEMS

1. Solve $\{ x + y = 5 ; xy = 6 \}$

ANSWER $(2, 3), (3, 2)$

2. Solve $\{ y = x + 1 ; x^2 + y^2 = 25 \}$

ANSWER $(3, 4), (-4, -3)$

3. Solve $\{ x - y = 1 ; xy = 12 \}$

ANSWER $(4, 3), (-3, -4)$

4. Solve $\{ y = x + 5 ; y = x^2 + 3 \}$

ANSWER $(2, 7), (-1, 4)$

5. Solve $\{ x + y = 7 ; x^2 + y^2 = 25 \}$

ANSWER $(3, 4), (4, 3)$

6. How many solutions has $\{ y = x + 6 ; y = x^2 \}$?

ANSWER Two

7. How many solutions has $\{ y = x - 3 ; y = x^2 \}$?

ANSWER None

Hard · stretch and reason

7 PROBLEMS

1. For which c is $y = x + c$ a tangent to $x^2 + y^2 = 8$?

ANSWER $c = \pm 4$

2. Solve $\{ x + y = 6 ; x^2 + y^2 = 20 \}$

ANSWER $(2, 4), (4, 2)$

3. Solve $\{ \frac{1}{x} + \frac{1}{y} = \frac{5}{6} ; x + y = 5 \}$

ANSWER $(2, 3), (3, 2)$

4. Two numbers have sum 12 and product 35: find them

ANSWER 5 and 7

5. A rectangle has perimeter 26 and area 40: find its sides

ANSWER 5 and 8

6. For which k has $\{ y = kx ; y = x^2 + 1 \}$ exactly one solution?

ANSWER $k = \pm 2$

7. Solve $\{ x^2 + y^2 = 10 ; x^2 - y^2 = 6 \}$

ANSWER $(\pm 2\sqrt{2}, \pm \sqrt{2})$

07 Homework

Homework — 5 tasks

Every answer is a list of ordered pairs, and every pair is checked in both equations.

- Solve $\{ x + y = 8 ; xy = 15 \}$.
- Solve $\{ y = 2x - 1 ; x^2 + y^2 = 25 \}$.
- Solve $\{ y = x + 2 ; y = x^2 \}$ and say what the answer means on a graph.
- For which c is $y = 2x + c$ a tangent to $y = x^2$?

5. Two numbers have sum 9 and sum of squares 45. Find them.

Methods of solving systems of equations

Substitution, addition, and the symmetric trick — one of the three fits every system.

LESSON 30–31

2 × 40 MIN

ALGEBRA 9, §14

IGX 6.2

LESSON CLOCK

13'

23'

22'

18'

4'

Choosing a method	13 min
Addition	23 min
Symmetric systems	22 min
Checking	18 min
Homework	4 min

LEARNING OBJECTIVES

- Choose between substitution and addition from the shape of the system.
- Solve a system of two second-degree equations by adding or subtracting.
- Use the substitution $u = x + y$, $v = xy$ on a symmetric system.
- Reject pairs that fail the original equations.

TEXTBOOK

National	pp. 72–76
Cambridge	Core & Extended, pp. 124–128
Workbook	Exercise 6.2

01

Explanation

Choosing a method

THE SYSTEM LOOKS LIKE	USE	BECAUSE
one equation linear	substitution	one unknown is easy to isolate
both have x^2 and y^2	addition or subtraction	a whole square cancels
unchanged when x and y swap	$u = x + y$, $v = xy$	it becomes linear in u , v
every term of the same degree	divide by y^2	it becomes a quadratic in $\frac{x}{y}$

LOOK BEFORE YOU COMPUTE

Ten seconds spent naming the shape saves a page of algebra. The four rows above cover almost every system in the Grade 9 course, and the wrong choice is what makes a question take twenty minutes instead of four.

Addition

When both equations contain x^2 and y^2 , adding or subtracting removes one of them entirely.

Example. $\{ x^2 + y^2 = 25 ; x^2 - y^2 = 7 \}$.

STEP	WORKING
add	$2x^2 = 32 \Rightarrow x^2 = 16$
subtract	$2y^2 = 18 \Rightarrow y^2 = 9$
solve	$x = \pm 4, y = \pm 3$
pair them	$(4,3), (4,-3), (-4,3), (-4,-3)$

FOUR SIGN CHOICES MEANS FOUR SOLUTIONS – UNLESS A THIRD CONDITION CUTS THEM DOWN

Here both equations involve only squares, so all four pairs work. If the system had contained $xy = 12$ instead, only the two with matching signs would survive. Always test the pairs against every equation.

Symmetric systems

A system is **symmetric** if swapping x and y leaves it unchanged. Then $u = x + y$ and $v = xy$ turn it into something much smaller.

$$x^2 + y^2 = u^2 - 2v \quad x^3 + y^3 = u^3 - 3uv \quad \frac{1}{x} + \frac{1}{y} = \frac{v}{u} \quad \dots \text{i.e.} \quad \frac{u}{v}$$

Example. $\{ x + y = 5 ; x^2 + y^2 = 13 \}$. Here $u = 5$ and $u^2 - 2v = 13$, so $25 - 2v = 13$ and $v = 6$.

Then x and y are the roots of $t^2 - ut + v = 0$, that is $t^2 - 5t + 6 = 0$: $t = 2, 3$. The solutions are $(2, 3)$ and $(3, 2)$.

VIETA, RUNNING BACKWARDS

Two numbers with sum u and product v are exactly the roots of $t^2 - ut + v = 0$. That single fact is what makes the substitution work, and it is worth knowing on its own — it answers every “two numbers whose sum and product are...” question in one line.

Checking

Squaring, multiplying by an unknown, or dividing can all introduce pairs that solve the new system but not the old one.

OPERATION	RISK	GUARD
squaring both sides	false pairs appear	substitute back
dividing by y	$y = 0$ is lost	treat $y = 0$ separately
multiplying by x	$x = 0$ may appear	substitute back

SUBSTITUTING INTO ONE EQUATION IS NOT A CHECK

A false pair usually satisfies one of the two. Test every candidate in **both** original equations, or the check proves nothing.

02

Worked examples

EXAMPLE 1 Solve $\{ x^2 + y^2 = 25 ; x^2 - y^2 = 7 \}$.

1

Add: $2x^2 = 32 \Rightarrow x = \pm 4$.

2

Subtract: $2y^2 = 18 \Rightarrow y = \pm 3$.

3

All four sign combinations satisfy both.
Only squares appear.

4

$(\pm 4, \pm 3)$ — four pairs.

ANSWER

$(4,3), (4,-3), (-4,3), (-4,-3)$

EXAMPLE 2 Solve $\{x + y = 5; x^2 + y^2 = 13\}$.

① $u = 5$, and $x^2 + y^2 = u^2 - 2v$.

② $25 - 2v = 13 \Rightarrow v = 6$

③ $t^2 - 5t + 6 = 0 \Rightarrow t = 2, 3$

④ $(2, 3)$ and $(3, 2)$.

ANSWER $(2, 3)$ and $(3, 2)$ **EXAMPLE 3** Solve $\{x^2 + xy = 12; xy + y^2 = 24\}$.

① Add: $x^2 + 2xy + y^2 = 36$.

② $(x + y)^2 = 36 \Rightarrow x + y = \pm 6$

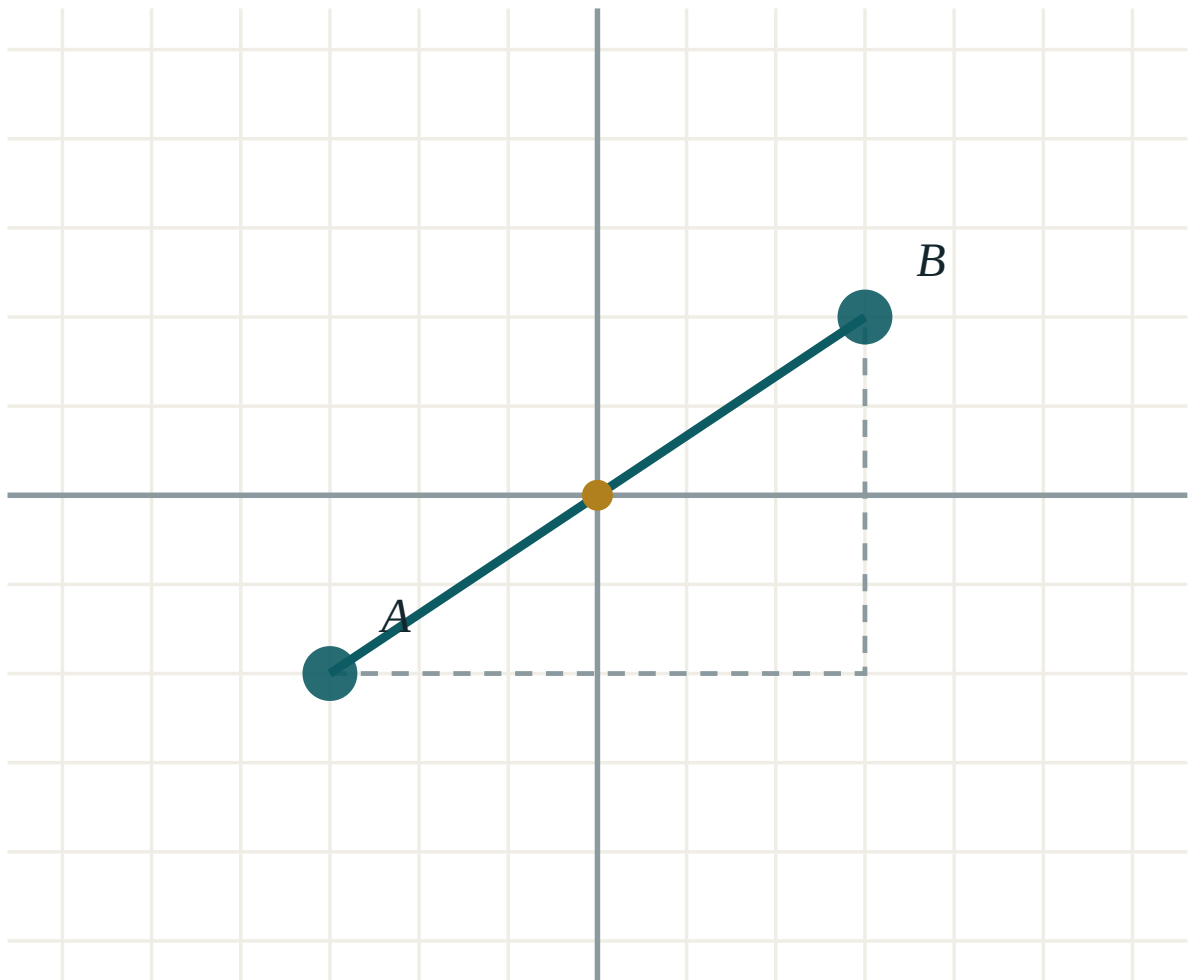
③ With $x + y = 6$: $x(x + y) = 12 \Rightarrow 6x = 12 \Rightarrow x = 2, y = 4$.

④ With $x + y = -6$: $x = -2, y = -4$.

ANSWER $(2, 4)$ and $(-2, -4)$ **03 Interactive model**

Write four systems on the board and ask only which method each needs — no solving — until every pupil names all four correctly.

Two curves, and where they meet



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Method of addition	Qo'shish usuli	Способ сложения
Elimination	Yo'qotish	Исключение
Symmetric system	Simmetrik sistema	Симметричная система
Auxiliary unknown	Yordamchi noma'lum	Вспомогательная переменная
Homogeneous system	Bir jinsli sistema	Однородная система
Equivalent systems	Teng kuchli sistemalar	Равносильные системы
Extraneous pair	Chet juftlik	Посторонняя пара
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Both equations have x^2 and y^2 . Use:

substitution

addition or subtraction

squaring

a graph

A symmetric system is best handled by:

$u = x + y, v = xy$

squaring

dividing

guessing

$x^2 + y^2$ in terms of u and v :

$u^2 - v$

$u^2 - 2v$

$u^2 + 2v$

$u - 2v$

Numbers with sum u and product v are roots of:

$$t^2 + ut + v = 0$$

$$t^2 - ut + v = 0$$

$$t^2 - ut - v = 0$$

$$t^2 + v = 0$$

Dividing by y risks losing:

$$y = 1$$

$$y = 0$$

$$x = 0$$

nothing

A candidate pair must be tested in:

one equation

both equations

neither

the simpler one

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Which method for $\{ y = 2x ; x^2 + y^2 = 20 \}$?

ANSWER Substitution

2. Which method for $\{ x^2 + y^2 = 25 ; x^2 - y^2 = 7 \}$?

ANSWER Addition

3. Which method for $\{ x + y = 5 ; xy = 6 \}$?

ANSWER Symmetric — u, v

4. $x^2 + y^2$ in terms of u, v

ANSWER $u^2 - 2v$

5. Roots of $t^2 - 5t + 6 = 0$

ANSWER 2 and 3

6. Solve $\{ x^2 = 16 ; y^2 = 9 \}$

ANSWER $(\pm 4, \pm 3)$

7. Solve $\{ x + y = 7 ; x - y = 1 \}$

ANSWER $(4, 3)$

Medium · the standard question

7 PROBLEMS

1. Solve $\{ x^2 + y^2 = 25 ; x^2 - y^2 = 7 \}$

ANSWER $(\pm 4, \pm 3)$ — four pairs

2. Solve $\{ x + y = 5 ; x^2 + y^2 = 13 \}$

ANSWER $(2, 3), (3, 2)$

3. Solve $\{ x + y = 6 ; xy = 8 \}$

ANSWER $(2, 4), (4, 2)$

4. Solve $\{ x^2 + y^2 = 10 ; xy = 3 \}$

ANSWER $(1, 3), (3, 1), (-1, -3), (-3, -1)$

5. Solve $\{ x^2 + xy = 12 ; xy + y^2 = 24 \}$

ANSWER $(2, 4), (-2, -4)$

6. Solve $\{ x - y = 2 ; x^2 - y^2 = 16 \}$

ANSWER $(5, 3)$

7. Two numbers: sum 10, product 21

ANSWER 3 and 7

Hard · stretch and reason

7 PROBLEMS

1. Solve $\{ x + y = 4 ; x^3 + y^3 = 28 \}$

ANSWER $(1, 3), (3, 1)$

2. Solve $\{ \frac{1}{x} + \frac{1}{y} = \frac{3}{4} ; xy = 8 \}$

ANSWER $(2, 4), (4, 2)$

3. Solve $\{ x^2 - 3xy + 2y^2 = 0 ; x + y = 6 \}$

ANSWER $(3, 3), (4, 2)$

4. Solve $\{ x^2 + y^2 = 5 ; x^2 - xy = 0 \}$

ANSWER $(0, \pm\sqrt{5})$ and $(\pm\frac{\sqrt{10}}{2}, \pm\frac{\sqrt{10}}{2})$

5. Two numbers: sum 6, sum of squares 26

ANSWER 1 and 5

6. A rectangle: perimeter 34, diagonal 13

ANSWER 5 by 12

7. Solve $\{ x + y + xy = 11 ; x + y - xy = -1 \}$

ANSWER $(2, 3), (3, 2)$

07 Homework

Homework — 5 tasks

Name the method before solving; the name is worth a mark on its own.

1. Solve $\{ x^2 + y^2 = 34 ; x^2 - y^2 = 16 \}$.
2. Solve $\{ x + y = 7 ; x^2 + y^2 = 29 \}$ using u and v .
3. Solve $\{ x + y = 9 ; xy = 20 \}$.
4. Two numbers have sum 11 and sum of squares 65. Find them.

5. Explain in two sentences why dividing a system by y can lose a solution.

Systems of second-degree inequalities with one unknown

Solve each line on its own number line, then keep only what every line agrees on.

LESSON 32–33

2 × 40 MIN

ALGEBRA 9, §15

IGX 14.3

LESSON CLOCK

11'

22'

20'

23'

4'

One line at a time	11 min
Two inequalities together	22 min
Systems and collections	20 min
Harder shapes	23 min
Homework	4 min

LEARNING OBJECTIVES

- Solve each inequality of a system separately by the sign of the parabola.
- Draw the solution sets on one number line and read off the intersection.
- Distinguish a system (and) from a collection (or).
- Write the answer in interval notation, with the right kind of bracket.

TEXTBOOK

National	pp. 77–81
Cambridge	Core & Extended, pp. 292–296
Workbook	Exercise 14.3

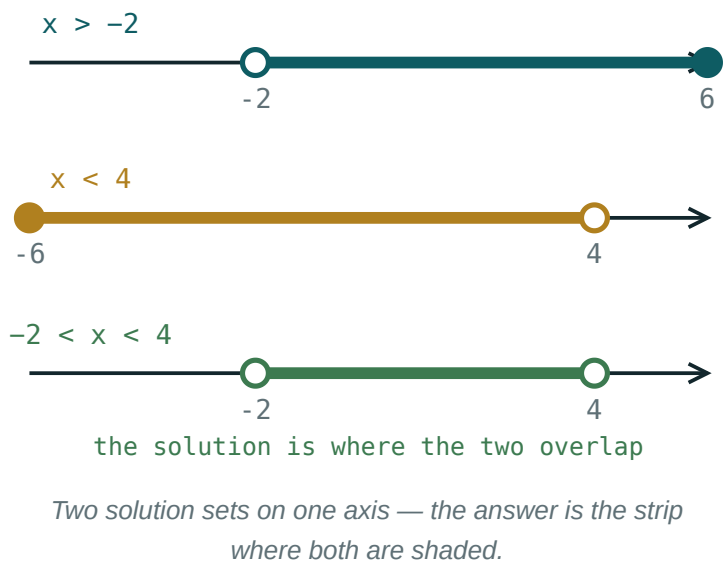
01

Explanation

One line at a time

A **system** of inequalities asks for the values of x that satisfy **every** line at once. There is no clever combined method: solve each line separately, then intersect.

STEP	WHAT YOU DO	WHY
1	bring each line to ... > 0 form	the sign rule needs one side zero
2	find the roots of each	they are the only places a sign can change
3	read the sign from a and the roots	outside for a 's sign, between for the other
4	draw all the solution sets on one axis	the intersection becomes visible
5	write the overlap	that is the answer



DRAW, DO NOT REASON IN WORDS

Almost every mark lost on this topic is lost by trying to combine $x < 5$ and $x > -2$ in the head. Two short strokes on a number line and the answer reads itself.

Two inequalities together

Example. $\{ x^2 - 4 > 0 ; x^2 - 9 < 0 \}$.

LINE	ROOTS	SOLUTION
$x^2 - 4 > 0$	$-2, 2$	$(-\infty, -2) \cup (2, +\infty)$
$x^2 - 9 < 0$	$-3, 3$	$(-3, 3)$
both	—	$(-3, -2) \cup (2, 3)$

The first is satisfied outside ± 2 because $a > 0$; the second inside ± 3 for the same reason read the other way. The overlap is two short intervals, not one.

AN ANSWER MAY BE TWO INTERVALS

Learners often assume the intersection of two sets is a single interval. It is whatever the picture shows — here two pieces, and sometimes $\emptyset$. Never force the answer into one bracket.

Systems and collections

Uzbek textbooks distinguish the two carefully, and so should the answer.

WRITTEN	NAME	MEANS	ANSWER IS
$\{ \dots ; \dots \}$	system	both hold	the intersection
$[\dots ; \dots]$	collection	at least one holds	the union

So $\{ x > 1 ; x > 4 \}$ gives $x > 4$, while $[x > 1 ; x > 4]$ gives $x > 1$. The same two lines, opposite answers.

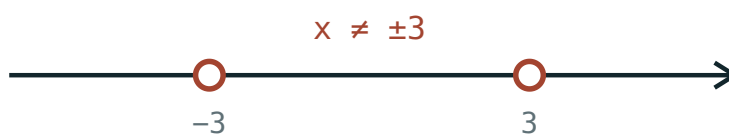
A DOUBLE INEQUALITY IS A SYSTEM IN DISGUISE

$-3 < x^2 - 4 < 5$ is exactly the system $\{ x^2 - 4 > -3 ; x^2 - 4 < 5 \}$. Split it, solve both, intersect. Trying to work on all three parts at once is where sign errors breed.

Harder shapes

Three cases worth practising before the control work:

- **No solution.** $\{ x^2 + 1 < 0 ; x > 0 \}$ — the first line is never true, so the answer is $\emptyset$. Say so; do not leave the page blank.
- **One point.** $\{ x^2 \leq 0 ; x \geq 0 \}$ gives $x = 0$ alone — a single number is a perfectly good answer set.
- **A fraction.** $\frac{x-1}{x+2} \geq 0$ is not a system, but it is solved the same way: mark 1 filled, -2 open, alternate the signs.



A filled dot is admitted, an open dot is not — the bracket follows the dot.

THE BRACKET FOLLOWS THE DOT, ALWAYS

$\geq$ on a numerator gives a square bracket; a zero of a **denominator** is always a round bracket, whatever the sign in the inequality. Mixing these up is the single most common slip in this chapter.

02 Worked examples

EXAMPLE 1 Solve $\{x^2 - 4 > 0; x^2 - 9 < 0\}$.

1 $x^2 - 4 > 0 \Rightarrow x < -2$ or $x > 2$.

Outside the roots.

2 $x^2 - 9 < 0 \Rightarrow -3 < x < 3$.

Between the roots.

3 Draw both on one axis.

4 Overlap: $(-3, -2) \cup (2, 3)$.

Two pieces.

ANSWER

$(-3, -2) \cup (2, 3)$

EXAMPLE 2 Solve $\{x^2 - 5x + 6 \leq 0; x - 1 > 0\}$.

1 $(x - 2)(x - 3) \leq 0 \Rightarrow 2 \leq x \leq 3$.

2 $x > 1$.

3 The whole of $[2, 3]$ lies beyond 1.

4 $[2, 3]$.

ANSWER

$[2, 3]$

EXAMPLE 3 Solve $\frac{x-1}{x+2} \geq 0$.

① Zeros: $x = 1$ (filled), $x = -2$ (open).

Denominator never filled.

② Three intervals; test one point in each.

③ $x = -3$: +, $x = 0$: -, $x = 2$: +.

④ $(-\infty, -2) \cup [1, +\infty)$.

ANSWER

$(-\infty, -2) \cup [1, +\infty)$

03 Interactive model

Give the class the same two lines twice — once as a system, once as a collection — and let them discover that the answers differ.

Two sets on one axis

$$x > 2$$



inequality $x > 2$

boundary 2 (open)

$x = 0$ works? no

$x = 6$ works? yes

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
System of inequalities	Tengsizliklar sistemasi	Система неравенств
Collection of inequalities	Tengsizliklar majmuasi	Совокупность неравенств
Intersection	Kesishma	Пересечение
Union	Birlashma	Объединение
Number line	Sonlar o'qi	Числовая прямая
Interval	Oraliq	Промежуток
Closed bracket	Yopiq qavs	Закрывающая скобка
Empty set	Bo'sh to'plam	Пустое множество

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A system of inequalities asks for:

the union

the intersection

either

neither

$\{ x > 1 ; x > 4 \}$ gives:

$x > 1$

$x > 4$

$1 < x < 4$

$\emptyset$

$[x > 1 ; x > 4]$ gives:

$x > 1$

$x > 4$

$1 < x < 4$

$\emptyset$

$x^2 - 9 < 0$ solves to:

$x < -3$ or $x > 3$

$-3 < x < 3$

$x < 3$

$\emptyset$

A zero of a denominator is drawn:

filled

open

either

twice

$\{ x^2 + 1 < 0 ; x > 0 \}$ has:

one solution

no solution

all x

$x > 0$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x^2 - 4 > 0$

ANSWER $(-\infty, -2) \cup (2, +\infty)$

2. Solve $x^2 - 9 < 0$

ANSWER $(-3, 3)$

3. Solve $\{ x > 2 ; x > 5 \}$

ANSWER $x > 5$

4. Solve $\{ x < 3 ; x > 0 \}$

ANSWER $(0, 3)$

5. Solve $[x < 0 ; x > 4]$

ANSWER $(-\infty, 0) \cup (4, +\infty)$

6. Solve $x^2 \leq 0$

ANSWER $x = 0$

7. Solve $x^2 + 4 > 0$

ANSWER all x

Medium · the standard question

7 PROBLEMS

1. Solve $\{x^2 - 4 > 0; x^2 - 9 < 0\}$

ANSWER $(-3, -2) \cup (2, 3)$

2. Solve $\{x^2 - 5x + 6 \leq 0; x > 1\}$

ANSWER $[2, 3]$

3. Solve $\{x^2 - x - 6 < 0; x \geq 0\}$

ANSWER $[0, 3)$

4. Solve $-3 < x^2 - 4 < 5$

ANSWER $(-3, -1) \cup (1, 3)$

5. Solve $\frac{x-1}{x+2} \geq 0$

ANSWER $(-\infty, -2) \cup [1, +\infty)$

6. Solve $\{x^2 > 1; x < 0\}$

ANSWER $(-\infty, -1)$

7. Solve $\{x^2 - 2x \leq 0; x \neq 1\}$

ANSWER $[0, 1) \cup (1, 2]$
Hard · stretch and reason

7 PROBLEMS

1. Solve $\{x^2 - 7x + 10 < 0; x^2 - 4x + 3 > 0\}$

ANSWER $(3, 5)$

2. Solve $\frac{x^2 - 4}{x - 1} \leq 0$

ANSWER $(-\infty, -2] \cup (1, 2]$

3. Solve $\{x^2 + x - 2 > 0; x^2 - 4x < 0\}$

ANSWER $(1, 4)$

4. For which a has $\{x^2 < 4; x > a\}$ no solution?

ANSWER $a \geq 2$

5. Solve $1 \leq x^2 \leq 4$

ANSWER $[-2, -1] \cup [1, 2]$

6. Solve $\{x^2 - 6x + 9 \leq 0; x < 5\}$

ANSWER $x = 3$

7. Solve $\{(x - 1)^2 > 0; x^2 - 4 \leq 0\}$

ANSWER $[-2, 1) \cup (1, 2]$

07 Homework

Homework — 5 tasks

Draw the number line for every task, even the ones you can see at once.

1. Solve $\{x^2 - 1 > 0; x^2 - 16 < 0\}$.

2. Solve $\{x^2 - 3x \leq 0; x > 1\}$.

3. Solve $-1 < x^2 - 5 < 4$.

4. Solve $\frac{x + 3}{x - 2} < 0$.

5. For which a does $\{ x^2 \leq 9 ; x \geq a \}$ have exactly one solution?

Proving the simplest inequalities

A square is never negative — nearly every proof in this section is that one fact, arranged.

LESSON 34–35

2 × 40 MIN

ALGEBRA 9, §16

IGX 14.4

LESSON CLOCK

9'

23'

22'

22'

4'

Proving is not solving	9 min
The square method	23 min
The mean inequality	22 min
Equality, and counter-examples	22 min
Homework	4 min

LEARNING OBJECTIVES

- Prove an inequality by moving everything to one side and completing a square.
- State and use the arithmetic–geometric mean inequality for two numbers.
- Say where equality holds, and prove that it holds only there.
- Distinguish proving an inequality from solving one.

TEXTBOOK

National	pp. 82–86
Cambridge	Core & Extended, pp. 297–300
Workbook	Exercise 14.4

01

Explanation

Proving is not solving

	SOLVING	PROVING
the question	for which x ?	for all x ?
the answer	a set	a chain of true statements
one value is	a solution	not a proof
one failing value is	nothing	a complete disproof

The asymmetry in the last row is worth dwelling on. To prove $x^2 + 1 \geq 2x$ you must argue for every x at once; to disprove $x^2 \geq x$ you need only exhibit $x = \frac{1}{2}$.

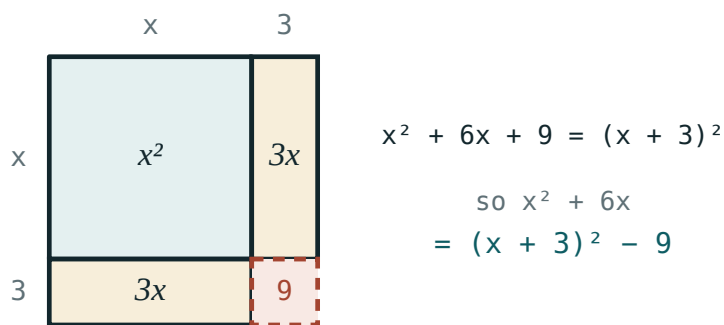
THE STANDARD OPENING

Move everything to one side. The inequality $A \geq B$ becomes $A - B \geq 0$, and now there is only one thing to do: show that the expression on the left cannot be negative.

The square method

The whole method rests on $t^2 \geq 0$ for every real t . Everything else is rearrangement.

TO PROVE	DIFFERENCE	WHY IT IS ≥ 0
$x^2 + 1 \geq 2x$	$(x - 1)^2$	a square
$a^2 + b^2 \geq 2ab$	$(a - b)^2$	a square
$x^2 + x + 1 > 0$	$(x + \frac{1}{2})^2 + \frac{3}{4}$	square plus a positive
$a^2 + b^2 + c^2 \geq ab + bc + ca$	$\frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$	three squares



Completing the square is what turns an unclear sign into an obvious one.

WRITE THE SQUARE, DO NOT DESCRIBE IT

“The left side is bigger because of the square” earns nothing. The proof is the line $x^2 + 1 - 2x = (x - 1)^2 \geq 0$ — the identity, then the reason, then the conclusion.

The mean inequality

For $a \geq 0$ and $b \geq 0$:

$$\frac{a + b}{2} \geq \sqrt{ab}$$

The proof is one line of the previous section. Since $(\sqrt{a} - \sqrt{b})^2 \geq 0$, expanding gives $a - 2\sqrt{ab} + b \geq 0$, that is $a + b \geq 2\sqrt{ab}$.

FORM	SAYS
$a + b \geq 2\sqrt{ab}$	the sum is at least twice the geometric mean
$x + \frac{1}{x} \geq 2 \ (x > 0)$	the classic special case, $b = \frac{1}{a}$
$(a + b)(\frac{1}{a} + \frac{1}{b}) \geq 4$	two applications, multiplied

THE RECTANGLE BEHIND IT

Of all rectangles with a given area, the square has the smallest perimeter. That is exactly $a + b \geq 2\sqrt{ab}$, with ab the area — and it is why the equality case is $a = b$.

Equality, and counter-examples

A complete answer names the equality case, and a complete disproof names a value.

STATEMENT	EQUALITY WHEN
$x^2 + 1 \geq 2x$	$x = 1$
$a + b \geq 2\sqrt{ab}$	$a = b$
$x + \frac{1}{x} \geq 2$	$x = 1$
$a^2 + b^2 + c^2 \geq ab + bc + ca$	$a = b = c$

And two statements that are simply false, with the value that kills them:

- “ $x^2 \geq x$ for all x ” — take $x = \frac{1}{2}$: $\frac{1}{4} < \frac{1}{2}$.
- “ $x + \frac{1}{x} \geq 2$ for all $x \neq 0$ ” — take $x = -1$: the sum is -2 .

THE CONDITION $x > 0$ IS PART OF THE STATEMENT

The second bullet is the reason. Dropping a condition because it looks decorative turns a theorem into a false claim — and in a proof, a dropped condition is a lost proof.

02 Worked examples

EXAMPLE 1 Prove $x^2 + 1 \geq 2x$ for all real x .

- 1 Move everything left: $x^2 - 2x + 1 \geq 0$.
- 2 Factorise: $(x - 1)^2 \geq 0$.
An identity.
- 3 A square is never negative — true for every x .
- 4 Equality only when $x = 1$.

ANSWER

Proved; equality at $x = 1$

EXAMPLE 2 Prove $a + b \geq 2\sqrt{ab}$ for $a, b \geq 0$.

- 1 Both roots exist because $a, b \geq 0$.
- 2 $(\sqrt{a} - \sqrt{b})^2 \geq 0$
- 3 Expand: $a - 2\sqrt{ab} + b \geq 0$.
- 4 Rearrange: $a + b \geq 2\sqrt{ab}$, equality at $a = b$.

ANSWER

Proved; equality at $a = b$

EXAMPLE 3 Prove $a^2 + b^2 + c^2 \geq ab + bc + ca$.

- ① Double both sides: $2a^2 + 2b^2 + 2c^2 \geq 2ab + 2bc + 2ca$.
Easier to split.
- ② Difference = $(a-b)^2 + (b-c)^2 + (c-a)^2$.
- ③ A sum of squares is non-negative.
- ④ Equality only when $a = b = c$.

ANSWERProved; equality at $a = b = c$ **03** Interactive model

Ask the class to disprove " $x^2 \geq x$ for all x " before proving anything — one fraction settles it, and the difference between proving and disproving lands immediately.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To prove	Isbotlash	Доказать
Identity	Ayniyat	Тождество
Non-negative	Manfiy emas	Неотрицательный
Perfect square	To'la kvadrat	Полный квадрат
Arithmetic mean	O'rta arifmetik	Среднее арифметическое
Geometric mean	O'rta geometrik	Среднее геометрическое
Equality case	Tenglik holati	Случай равенства
Counter-example	Qarama-qarshi misol	Контрпример

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To prove $A \geq B$ you show:

$$A - B \geq 0$$

$$A > 0$$

$$B < 0$$

one value works

The engine of nearly every proof here is:

$$t^2 \geq 0$$

$$t > 0$$

$$t \neq 0$$

$$t = 0$$

Equality in $x^2 + 1 \geq 2x$ holds at:

$$x = 0$$

$$x = 1$$

$$x = -1$$

never

$a + b \geq 2\sqrt{ab}$ needs:

nothing

$a, b \geq 0$

$a > b$

$a = b$

One value that fails a claim is:

nothing

a full disproof

a partial proof

a proof

$x + \frac{1}{x} \geq 2$ requires:

$x \neq 0$

$x > 0$

$x < 0$

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Prove $x^2 \geq 0$

ANSWER A square is non-negative

2. Prove $x^2 + 4 > 0$

ANSWER $x^2 \geq 0$, add 4

3. Prove $(x - 3)^2 \geq 0$

ANSWER A square

4. Prove $a^2 + b^2 \geq 2ab$

ANSWER $(a - b)^2 \geq 0$

5. Prove $x^2 + 1 \geq 2x$

ANSWER $(x - 1)^2 \geq 0$

6. Disprove “ $x^2 \geq x$ for all x ”

ANSWER $x = \frac{1}{2}$

7. Equality in $a + b \geq 2\sqrt{ab}$

ANSWER $a = b$

Medium · the standard question

7 PROBLEMS

1. Prove $x^2 + x + 1 > 0$

ANSWER $(x + \frac{1}{2})^2 + \frac{3}{4} > 0$

2. Prove $x + \frac{1}{x} \geq 2$ for $x > 0$

ANSWER $(\sqrt{x} - \frac{1}{\sqrt{x}})^2 \geq 0$

3. Prove $a^2 + b^2 + c^2 \geq ab + bc + ca$

ANSWER Three squares, halved

4. Prove $(a + b)^2 \geq 4ab$

ANSWER $(a - b)^2 \geq 0$

5. Prove $a^4 + b^4 \geq a^3b + ab^3$

ANSWER $(a - b)^2(a^2 + ab + b^2) \geq 0$

6. Prove $x^4 + 1 \geq x^3 + x$ for $x > 0$

ANSWER $(x^3 - 1)(x - 1) \geq 0$

7. Disprove " $a + b \geq 2\sqrt{ab}$ for all reals"

ANSWER $a = b = -1$

Hard · stretch and reason

7 PROBLEMS

1. Prove $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$ for $a, b > 0$

ANSWER Multiply two mean inequalities

2. Prove $a^2 + b^2 + 1 \geq ab + a + b$

ANSWER Half a sum of three squares

3. Least value of $x + \frac{4}{x}$ for $x > 0$

ANSWER 4, at $x = 2$

4. Prove $\frac{a}{b} + \frac{b}{a} \geq 2$ for $ab > 0$

ANSWER $(a - b)^2 \geq 0$, divided by ab

5. Prove $a^3 + b^3 \geq ab(a + b)$ for $a, b > 0$

ANSWER $(a + b)(a - b)^2 \geq 0$

6. Greatest area of a rectangle of perimeter 20

ANSWER 25 — the square

7. Prove $(1 + a)(1 + b) \geq (1 + \sqrt{ab})^2$ for $a, b \geq 0$

ANSWER Expand; reduces to $(\sqrt{a} - \sqrt{b})^2 \geq 0$

07 Homework

Homework — 5 tasks

Every proof ends with the equality case named; a proof without it is incomplete.

1. Prove $x^2 + 9 \geq 6x$ and say when equality holds.
2. Prove $a^2 + b^2 \geq 2ab$ and interpret it as a statement about rectangles.
3. Prove $x + \frac{9}{x} \geq 6$ for $x > 0$.

4. Disprove " $x^3 \geq x^2$ for all x ".
5. Prove $a^2 + b^2 + c^2 \geq ab + bc + ca$ and name the equality case.

Chapter exercises — sets, Venn diagrams and notation

Cambridge writes solution sets in the language of sets; this hour makes both notations one.

LESSON 36

1 × 40 MIN

ALGEBRA 9, II BOB MASHQLARI

IGX 1.1–1.2, 14.1

LESSON CLOCK

8'

12'

12'

6'

2'

The symbols	8 min
Venn diagrams	12 min
Sets and intervals	12 min
Mixed exercises	6 min
Homework	2 min

LEARNING OBJECTIVES

- Read and write the set notation $\in$, $\subset$, $\cap$, $\cup$, $\emptyset$ and ξ .
- Show two or three sets on a Venn diagram and shade a described region.
- Translate an interval answer into set-builder notation and back.
- Recognise a system as an intersection and a collection as a union.

TEXTBOOK

National	pp. 87–90
Cambridge	Core & Extended, pp. 2–14
Workbook	Exercise 1.1–1.2

01

Explanation

The symbols

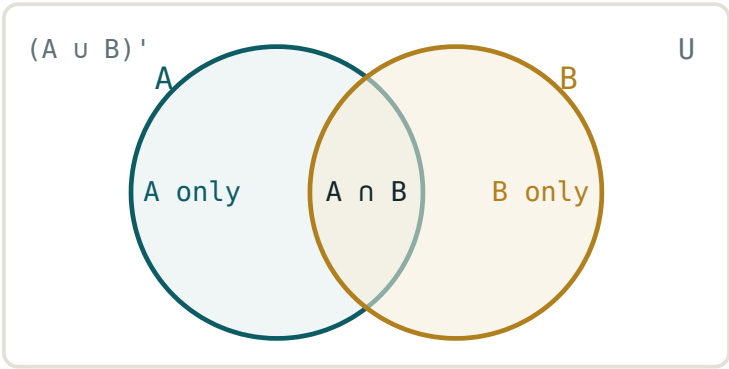
SYMBOL	READ AS	EXAMPLE
$\in$	is an element of	$3 \in \mathbb{N}$
$\notin$	is not an element of	$\sqrt{2} \notin \mathbb{Q}$
$\subset$	is a subset of	$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$
$\cap$	intersection — “and”	$A \cap B$
$\cup$	union — “or”	$A \cup B$
$\emptyset$	the empty set	$\{x : x^2 < 0\} = \emptyset$
ξ	the universal set	everything under discussion
A'	the complement of A	everything in ξ not in A
$n(A)$	the number of elements	$n(\{2,4,6\}) = 3$

ONE IDEA, TWO ALPHABETS

A **system** of inequalities is an $\cap$; a **collection** is a $\cup$. The Uzbek textbook writes the braces, Cambridge writes the symbols. Nothing mathematical separates them.

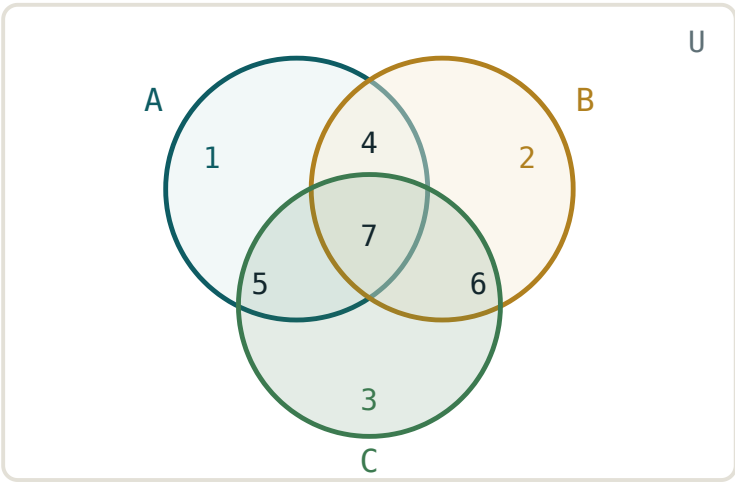
Venn diagrams

Two sets divide the universal set into four regions; three sets into eight.



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Two sets, four regions — both, only A, only B, neither.



seven regions inside, one outside

Three sets, eight regions — the centre belongs to all three.

REGION	NOTATION
in both	$A \cap B$
in A only	$A \cap B'$
in at least one	$A \cup B$
in neither	$(A \cup B)'$

Counting follows the picture: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$, because the overlap would otherwise be counted twice.

Sets and intervals

Every answer from the last two lessons can be written either way.

INTERVAL	SET-BUILDER
$(-3, 3)$	$\{x \in \mathbb{R} : -3 < x < 3\}$
$[2, 3]$	$\{x \in \mathbb{R} : 2 \leq x \leq 3\}$
$(-\infty, -2) \cup (2, +\infty)$	$\{x \in \mathbb{R} : x^2 > 4\}$
$\emptyset$	$\{x \in \mathbb{R} : x^2 + 1 < 0\}$

CAMBRIDGE MARKS THE NOTATION, NOT ONLY THE ANSWER

A correct set written with the wrong brackets loses a mark in an IGCSE paper. Round for excluded, square for included, and the colon inside set-builder braces reads “such that”.

02 Worked examples

EXAMPLE 1 With $\xi = \{1, \dots, 10\}$, $A = \{\text{even}\}$, $B = \{\text{multiples of } 3\}$, list $A \cap B$ and $A \cup B$.

① $A = \{2, 4, 6, 8, 10\}$

② $B = \{3, 6, 9\}$

③ $A \cap B = \{6\}$

In both.

④ $A \cup B = \{2, 3, 4, 6, 8, 9, 10\}$

Eight elements.

ANSWER

$\{6\}$ and $\{2, 3, 4, 6, 8, 9, 10\}$

EXAMPLE 2 In a class of 30, 18 play football, 14 play chess, 6 play both. How many play neither?

① $n(F \cup C) = 18 + 14 - 6 = 26$

Overlap counted once.

② $30 - 26 = 4$

③ Check the four regions: $12 + 6 + 8 + 4 = 30 \checkmark$

④ 4 pupils.

ANSWER

4

EXAMPLE 3 Write the solution of $\{x^2 - 4 > 0; x^2 - 9 < 0\}$ in set notation.

- 1 First set: $\{x : x < -2\} \cup \{x : x > 2\}$.
- 2 Second set: $\{x : -3 < x < 3\}$.
- 3 A system is an intersection.
- 4 $\{x \in \mathbb{R} : -3 < x < -2\} \cup \{x \in \mathbb{R} : 2 < x < 3\}$

ANSWER

$(-3, -2) \cup (2, 3)$

03 Interactive model

Shade the four regions of a two-set Venn diagram on the board and let the class name each one in symbols before any counting begins.

Set notation at speed

$3 \in A$ means:

3 is an element of A

A is a subset of 3

3 equals A

3 is not in A

$A \subset B$ means:

every element of A is in B

they share one element

they are equal

B is inside A

$A \cap B$ is:

everything in both

everything in either

what is only in A

the empty set

$A \cup B$ is:

everything in both

everything in either or both

what is in neither

the difference

If $A = \{1, 2, 3\}$ and $B = \{3, 4\}$ then $A \cap B$ is:

$\{3\}$

$\{1, 2, 3, 4\}$

$\emptyset$

$\{1, 2\}$

With the same sets, $n(A \cup B)$ is:

4

5

6

7

The empty set is written:

$\{0\}$

$\emptyset$

0

$\{\emptyset\}$

A' means:

the subsets of A

everything in the universal set not in A

A doubled

the empty set

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Set	To'plam	Множество
Element	Element	Элемент
Subset	Qism to'plam	Подмножество
Universal set	Universal to'plam	Универсальное множество
Complement	To'ldiruvchi	Дополнение
Venn diagram	Venn diagrammasi	Диаграмма Венна
Set-builder notation	To'plamni yozish usuli	Способ задания множества
Disjoint	Kesishmaydigan	Непересекающиеся

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\cap$ means:

or

and

not

all

A system of inequalities corresponds to:

$\cup$

$\cap$

$\emptyset$

$\subset$

$n(A \cup B)$ equals:

$n(A) + n(B)$

$n(A) + n(B) - n(A \cap B)$

$n(A) - n(B)$

$n(A \cap B)$

A' is:

everything in A

everything not in A

$A \cap B$

$\emptyset$

$\{x : x^2 + 1 < 0\}$ is:

$\mathbb{R}$

$\emptyset$

$\{0\}$

$\{-1\}$

Three sets divide ξ into:

six regions

eight regions

four regions

nine regions

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is $5 \in \mathbb{N}$?

ANSWER Yes

2. Is $\sqrt{2} \in \mathbb{Q}$?

ANSWER No

3. $\{1,2,3\} \cap \{3,4\}$

ANSWER $\{3\}$

4. $\{1,2\} \cup \{2,3\}$

ANSWER $\{1,2,3\}$

5. $n(\{2,4,6,8\})$

ANSWER 4

6. Write $(1, 5)$ in set-builder form

ANSWER $\{x : 1 < x < 5\}$

7. Write $\{x : x \geq 0\}$ as an interval

ANSWER $[0, +\infty)$

Medium · the standard question

7 PROBLEMS

1. $\xi = \{1..10\}$, $A = \text{even}$, $B = \text{multiples of 3}$: $A \cap B$

ANSWER $\{6\}$

2. Same sets: $A \cup B$

ANSWER $\{2,3,4,6,8,9,10\}$

3. Same sets: A'

ANSWER $\{1,3,5,7,9\}$

4. $n(A) = 18$, $n(B) = 14$, $n(A \cap B) = 6$: $n(A \cup B)$

ANSWER 26

5. Of 30, how many in neither?

ANSWER 4

6. Write $(-\infty, -2) \cup (2, \infty)$ as one condition

ANSWER $\{x : x^2 > 4\}$

7. Shade “in A only”

ANSWER $A \cap B'$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\{x^2 < 9; x > 0\}$ in set notation

ANSWER $\{x \in \mathbb{R} : 0 < x < 3\}$

2. $n(\xi) = 40, n(A) = 25, n(B) = 20, n((A \cup B)') = 5: n(A \cap B)$

ANSWER 10

3. How many subsets has $\{a, b, c\}$?

ANSWER 8

4. Describe the shaded centre of three circles

ANSWER $A \cap B \cap C$

5. $A \subset B$ — simplify $A \cap B$

ANSWER A

6. $A \subset B$ — simplify $A \cup B$

ANSWER B

7. Of 50: 30 study French, 25 German, 8 neither. Both?

ANSWER 13

07 Homework

Homework — 5 tasks

Answer every task twice: once in interval notation, once in set-builder notation.

- With $\xi = \{1, \dots, 12\}$, list $A \cap B$ for A = multiples of 3 and B = even.
- In a group of 40, 22 play tennis, 18 swim, 7 do both. How many do neither?
- Write the solution of $x^2 - 16 \leq 0$ in both notations.
- Draw a three-set Venn diagram and shade $A \cap B \cap C'$.
- How many subsets does a set of 5 elements have?

Control work 3, and work on the mistakes

Systems and proofs in one paper, and Chapter II closed before trigonometry opens.

LESSON 37–38

2 × 40 MIN

ALGEBRA 9, NAZORAT ISHI 3

IGX 6, 14 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Solve a system of two equations by the method the system asks for.
- Solve a system of second-degree inequalities and write the answer as intervals.
- Prove one inequality in full, with the equality case named.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 68–90
Cambridge	Core & Extended, pp. 118–300
Workbook	Control paper A3

01

Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Solve $\{ x + y = 7 ; xy = 12 \}$	5	L28–29
2	Solve $\{ x^2 + y^2 = 20 ; x^2 - y^2 = 12 \}$	5	L30–31
3	Solve $\{ x^2 - 1 > 0 ; x^2 - 25 < 0 \}$	5	L32–33
4	Solve $\frac{x - 2}{x + 1} \leq 0$	5	L32–33
5	Prove $x^2 + 4 \geq 4x$, naming the equality case	5	L34–35
6	Write the answer to Q3 in set-builder notation	5	L36

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for checking the pair in **both** equations; Q2 one for pairing the signs correctly; Q3 one for a drawn number line; Q4 one for the round bracket at -1 ; Q5 one for the equality case. Five of the thirty marks are habits, again.

Naming the slip

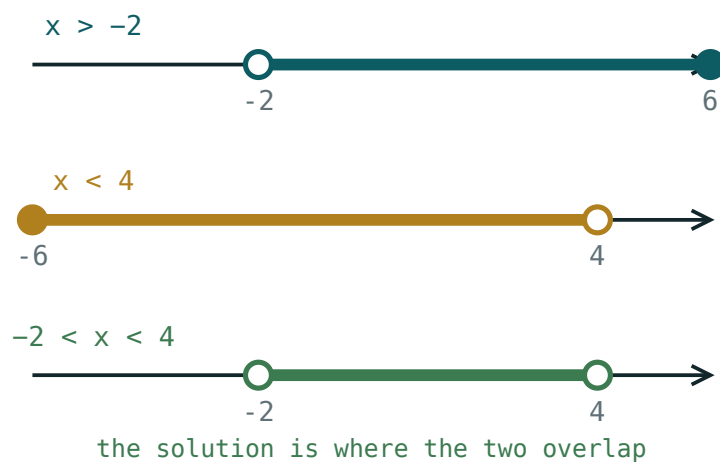
SLIP	WHAT IT LOOKS LIKE	THE FIX
one pair only	$(3, 4)$	the system is symmetric — $(4, 3)$ too
signs mispaired	$(4, 2)$	test in the second equation
intersection forced into one interval	$(-5, 5)$	the picture shows two pieces
square bracket on a denominator	$[-1, 2]$	$(-1, 2]$
proof by example	“try $x = 3$: true”	move to one side and factorise
equality case missing	$(x - 2)^2 \geq 0$	“...with equality at $x = 2$ ”
system read as collection	U instead of $\cap$	braces mean “and”

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter II as one map

Five boxes, and the links written as sentences:

- **a linear equation present** → **substitute** — “isolate, replace, solve one unknown”
- **two squares present** → **add or subtract** — “one square disappears”
- **symmetric** → $u = x + y, v = xy$ — “and then Vieta backwards”
- **a system of inequalities** → **one number line** — “solve separately, intersect”
- **an inequality to prove** → **one side, a square** — “and name the equality case”



Two sets, one axis — the whole of the inequality half of this chapter.

LOOKING FORWARD

Chapter III leaves algebra for a while: the angle escapes the triangle, the radian arrives, and sine and cosine become functions of any angle at all. Nothing in it needs Chapter II — but the equations it produces will be solved by exactly the methods just tested.

02 Worked examples

EXAMPLE 1 Model answer, Q1: solve $\{x + y = 7; xy = 12\}$.

- 1 Symmetric: $u = 7, v = 12$.
- 2 $t^2 - 7t + 12 = 0$
Vieta backwards.
- 3 $t = 3, 4$
- 4 $(3, 4)$ and $(4, 3)$ — both checked.

ANSWER

$(3, 4)$ and $(4, 3)$

EXAMPLE 2 Model answer, Q3: solve $\{x^2 - 1 > 0; x^2 - 25 < 0\}$.

① $x < -1$ or $x > 1$.
Outside ± 1 .

② $-5 < x < 5$.
Inside ± 5 .

③ One number line, both sets drawn.

④ $(-5, -1) \cup (1, 5)$.
Two pieces.

ANSWER

$(-5, -1) \cup (1, 5)$

EXAMPLE 3 Model answer, Q5: prove $x^2 + 4 \geq 4x$.

① $x^2 - 4x + 4 \geq 0$
Everything to one side.

② $(x - 2)^2 \geq 0$
An identity.

③ A square is never negative — true for all x .

④ Equality only at $x = 2$.

ANSWER

Proved; equality at $x = 2$

03 Interactive model

Mark Q5 in front of the class: read out a “proof” that only tests three values, and let them say what is missing.

Chapter II in twelve questions

A system with one linear equation is best solved by:

substitution

addition

squaring

a graph

Both equations have x^2 and y^2 : use

substitution

addition

Vieta

guessing

$x^2 + y^2$ in terms of u, v :

$u^2 - v$

$u^2 - 2v$

$u^2 + 2v$

$u + v$

Numbers with sum u , product v are roots of:

$$t^2 + ut + v = 0$$

$$t^2 - ut + v = 0$$

$$t^2 - v = 0$$

$$t^2 + v = 0$$

A candidate pair must be tested in:

one equation

both

neither

the easier one

A system of inequalities gives:

a union

an intersection

either

always one interval

$x^2 - 25 < 0$ solves to:

$$x < -5 \text{ or } x > 5$$

$$-5 < x < 5$$

$$x < 5$$

$$\emptyset$$

A zero of a denominator gets:

a square bracket

a round bracket

either

no bracket

To prove $A \geq B$:

test values

show $A - B \geq 0$

draw a graph

show $A > 0$

The engine of these proofs is:

$t^2 \geq 0$

$t > 0$

$t \neq 0$

$t = 1$

One failing value is:

nothing

a disproof

a proof

a special case

$\cap$ corresponds to:

a collection

a system

a complement

the empty set

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
System	Sistema	Система
Collection	Majmua	Совокупность
Substitution	O'rniga qo'yish	Подстановка
Method of addition	Qo'shish usuli	Способ сложения
Proof	Isbot	Доказательство
Equality case	Tenglik holati	Случай равенства
Interval notation	Oraliq belgisi	Обозначение промежутка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 is worth full marks only if you give:

one pair

both pairs

the sum

the product

Q3's answer has:

one interval

two intervals

no solution

one point

The bracket at -1 in Q4 is:

square

round

either

omitted

Q5 needs, besides the square:

a graph

the equality case

three test values

nothing

Q6 tests:

algebra

notation

speed

proof

Work on the mistakes means:

fix the wrong line

rewrite the solution

copy the answer

skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\{ x + y = 7 ; xy = 12 \}$

ANSWER $(3, 4), (4, 3)$

2. Solve $\{ x^2 = 9 ; y^2 = 4 \}$

ANSWER $(\pm 3, \pm 2)$

3. Solve $x^2 - 1 > 0$

ANSWER $(-\infty, -1) \cup (1, +\infty)$

4. Solve $x^2 - 25 < 0$

ANSWER $(-5, 5)$

5. Prove $x^2 + 4 \geq 4x$

ANSWER $(x - 2)^2 \geq 0$

6. Equality in $x^2 + 4 \geq 4x$

ANSWER $x = 2$

7. Write $(1, 5)$ in set-builder form

ANSWER $\{x : 1 < x < 5\}$

Medium · the standard question

7 PROBLEMS

1. Solve $\{ x^2 + y^2 = 20 ; x^2 - y^2 = 12 \}$

ANSWER $(\pm 4, \pm 2)$

2. Solve $\{ x^2 - 1 > 0 ; x^2 - 25 < 0 \}$

ANSWER $(-5, -1) \cup (1, 5)$

3. Solve $\frac{x-2}{x+1} \leq 0$

ANSWER $(-1, 2]$

4. Solve $\{ y = x + 1 ; x^2 + y^2 = 25 \}$

ANSWER $(3, 4), (-4, -3)$

5. Prove $a^2 + 9 \geq 6a$

ANSWER $(a - 3)^2 \geq 0$

6. Solve $\{ x + y = 6 ; x^2 + y^2 = 20 \}$

ANSWER $(2, 4), (4, 2)$

7. Write the answer to $x^2 \leq 4$ in set notation

ANSWER $\{x \in \mathbb{R} : -2 \leq x \leq 2\}$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\{x^2 + xy = 10; xy + y^2 = 15\}$

ANSWER $(2, 3), (-2, -3)$

2. Solve $\{x^2 - 4x + 3 < 0; x^2 - 9 > 0\}$

ANSWER $\emptyset$

3. Prove $a + \frac{1}{a} \geq 2$ for $a > 0$

ANSWER $(\sqrt{a} - \frac{1}{\sqrt{a}})^2 \geq 0$

4. For which k is $y = x + k$ tangent to $x^2 + y^2 = 8$?

ANSWER $k = \pm 4$

5. Solve $2 \leq x^2 - 2 \leq 7$

ANSWER $[-3, -2] \cup [2, 3]$

6. Two numbers: sum 8, sum of squares 34

ANSWER 3 and 5

7. Prove $(a + b)^2 \geq 4ab$ and name the equality case

ANSWER $(a - b)^2 \geq 0; a = b$

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on, in full, before starting the new chapter.

1. Solve $\{x + y = 9; xy = 20\}$.
2. Solve $\{x^2 + y^2 = 34; x^2 - y^2 = 16\}$.
3. Solve $\{x^2 - 4 > 0; x^2 - 36 < 0\}$ and write the answer in set notation.
4. Solve $\frac{x+4}{x-3} \geq 0$.

5. Prove $x^2 + 25 \geq 10x$ and name the equality case.

The radian measure of an angle

A second unit for angle — one that measures the arc rather than counting Babylonian steps.

LESSON 39–40

2 × 40 MIN

ALGEBRA 9, §17

IGX 15.1

LESSON CLOCK

13'

20'

20'

23'

4'

What a radian is	13 min
Converting	20 min
The angles worth knowing	20 min
Arcs and sectors	23 min
Homework	4 min

LEARNING OBJECTIVES

- Define one radian as the angle subtending an arc equal to the radius.
- Convert between degrees and radians in both directions.
- Know the common angles in both units without a table.
- Use $l = r\alpha$ and $S = \frac{1}{2}r^2\alpha$ for an arc and a sector.

TEXTBOOK

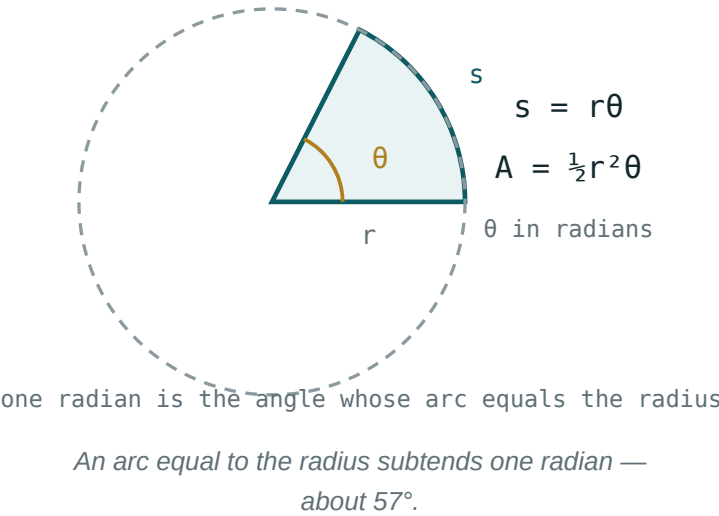
National	pp. 91–96
Cambridge	Core & Extended, pp. 304–309
Workbook	Exercise 15.1

01

Explanation

What a radian is

Take a circle of radius r and lay an arc of length r along it. The central angle you have made is **one radian**.



Nothing in that definition mentions the size of the circle, which is the point: a radian is a ratio of two lengths, so it is the same angle in every circle.

$$\alpha = \frac{l}{r}$$

— the angle, in radians, is arc ÷ radius

Going all the way round gives an arc of $2\pi r$, so a full turn is 2π radians. A straight angle is π , a right angle $\frac{\pi}{2}$.

WHY BOTHER WITH A SECOND UNIT

The 360 degrees of a full turn are a Babylonian convention — a good one, since 360 has many divisors, but arbitrary. The radian is not chosen; it falls out of the circle itself. Every formula in later mathematics that involves angles is simpler in radians, and from Grade 10 onwards degrees appear only in geometry questions.

Converting

One equation carries everything:

$$180^\circ = \pi \text{ rad}$$

DIRECTION	MULTIPLY BY	EXAMPLE
degrees → radians	$\frac{\pi}{180^\circ}$	$60^\circ = 60 \cdot \frac{\pi}{180} = \frac{\pi}{3}$
radians → degrees	$\frac{180^\circ}{\pi}$	$\frac{3\pi}{4} = \frac{3}{4} \cdot 180^\circ = 135^\circ$

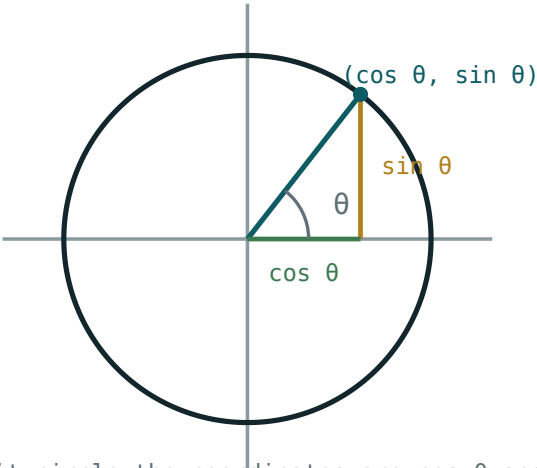
Radians are usually written as multiples of π and the word “rad” is left off: $x = \frac{\pi}{6}$ means an angle, not a number of degrees. A radian measure without π in it is a plain real number — $1 \text{ rad} \approx 57.3^\circ$.

SET THE CALCULATOR, OR EVERY ANSWER IS WRONG

$\sin 30 = 0.5$ in degree mode and ≈ -0.988 in radian mode. Neither is a mistake by the calculator. Check the DEG/RAD indicator before the first line of any trigonometry paper.

The angles worth knowing

DEGREES	0°	30°	45°	60°	90°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π



on the unit circle the coordinates are $\cos \theta$ and $\sin \theta$

The circle marked in both units — the eight angles that must be automatic.

LEARN THE ROW, NOT THE RULE

Converting 45° from first principles takes fifteen seconds; recognising $\frac{\pi}{4}$ takes none. Every question in the next four lessons uses these eight values, so the table above is worth memorising outright — read it left to right, then right to left, until both directions are instant.

Arcs and sectors

With α in **radians**, the two formulae are as short as they can be:

$$l = r\alpha \quad S = \frac{1}{2}r^2\alpha$$

QUANTITY	RADIANS	DEGREES
arc length	$l = r\alpha$	$l = \frac{\theta}{360} \cdot 2\pi r$
sector area	$S = \frac{1}{2}r^2\alpha$	$S = \frac{\theta}{360} \cdot \pi r^2$

Example. $r = 6 \text{ cm}$, $\alpha = \frac{\pi}{3}$. Then $l = 6 \cdot \frac{\pi}{3} = 2\pi \text{ cm}$ and $S = \frac{1}{2} \cdot 36 \cdot \frac{\pi}{3} = 6\pi \text{ cm}^2$.

$L = RA$ IS FALSE IN DEGREES

The compact formulae are the reward for using radians, and they are only true in radians. If the angle is given in degrees, convert first — or use the $\frac{\theta}{360}$ versions instead.

02 Worked examples

EXAMPLE 1 Convert 135° to radians and $\frac{5\pi}{6}$ to degrees.

$$1 \quad 135 \cdot \frac{\pi}{180} = \frac{135\pi}{180}$$

$$2 \quad \text{Cancel by 45: } \frac{3\pi}{4}.$$

$$3 \quad \frac{5\pi}{6} \cdot \frac{180}{\pi} = \frac{5 \cdot 180}{6}$$

$$4 \quad = 150^\circ$$

ANSWER

$$\frac{3\pi}{4} \text{ and } 150^\circ$$

EXAMPLE 2 A circle has $r = 6$ cm. Find the arc and sector for $\alpha = \frac{\pi}{3}$.

$$① \quad l = r\alpha = 6 \cdot \frac{\pi}{3}$$

$$② \quad l = 2\pi \approx 6.28 \text{ cm}$$

$$③ \quad S = \frac{1}{2} \cdot 6^2 \cdot \frac{\pi}{3} = \frac{1}{2} \cdot 36 \cdot \frac{\pi}{3}$$

$$④ \quad S = 6\pi \approx 18.85 \text{ cm}^2$$

ANSWER

$$l = 2\pi \text{ cm}, S = 6\pi \text{ cm}^2$$

EXAMPLE 3 An arc of 10 cm subtends 2 rad. Find the radius and the sector area.

$$① \quad \alpha = \frac{l}{r} \Rightarrow r = \frac{l}{\alpha}$$

$$② \quad r = \frac{10}{2} = 5 \text{ cm}$$

$$③ \quad S = \frac{1}{2} \cdot 25 \cdot 2$$

$$④ \quad S = 25 \text{ cm}^2$$

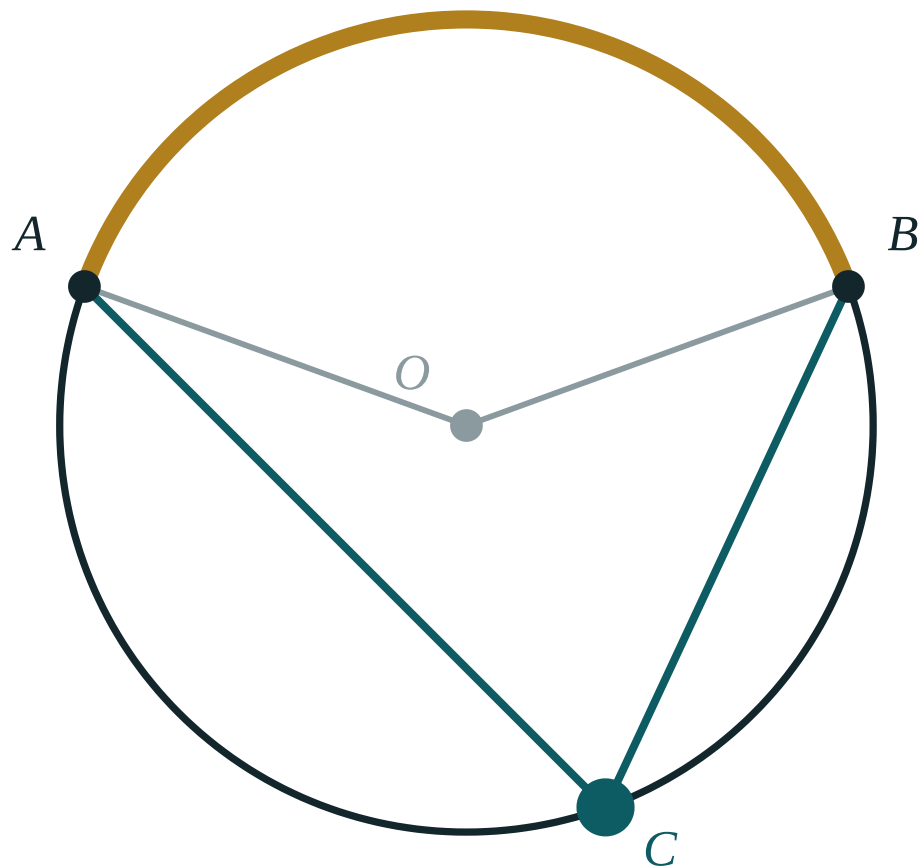
ANSWER

$$r = 5 \text{ cm}, S = 25 \text{ cm}^2$$

03 Interactive model

Cut a length of string equal to the radius of a large circle drawn on the board and bend it round the rim — one radian becomes a physical object, not a definition.

Degrees and radians on one circle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Radian	Radian	Радиян
Degree	Gradus	Градус
Arc	Yoy	Дуга
Arc length	Yoy uzunligi	Длина дуги
Sector	Sektor	Сектор
Central angle	Markaziy burchak	Центральный угол
Circumference	Aylana uzunligi	Длина окружности
Conversion	O‘tkazish	Перевод

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

One radian is the angle whose arc equals:

the diameter

the radius

the circumference

π

180° in radians:

$\frac{\pi}{2}$

π

2π

$\frac{\pi}{4}$

60° in radians:

$$\frac{\pi}{6}$$

$$\frac{\pi}{4}$$

$$\frac{\pi}{3}$$

$$\frac{\pi}{2}$$

$\frac{3\pi}{4}$ in degrees:

$$120^\circ$$

$$135^\circ$$

$$150^\circ$$

$$45^\circ$$

Arc length, α in radians:

$$l = r\alpha$$

$$l = r^2\alpha$$

$$l = \frac{\alpha}{r}$$

$$l = 2\pi r\alpha$$

Sector area, α in radians:

$r^2\alpha$

$\frac{1}{2}r^2\alpha$

$\frac{1}{2}r\alpha$

$\pi r^2\alpha$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 90° in radians

ANSWER $\frac{\pi}{2}$

2. 45° in radians

ANSWER $\frac{\pi}{4}$

3. 30° in radians

ANSWER $\frac{\pi}{6}$

4. π in degrees

ANSWER 180°

5. $\frac{\pi}{3}$ in degrees

ANSWER 60°

6. $r = 4, \alpha = 2$: arc

ANSWER 8

7. $r = 2, \alpha = 3$: sector area

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. 135° in radians

ANSWER $\frac{3\pi}{4}$

2. 210° in radians

ANSWER $\frac{7\pi}{6}$

3. $\frac{5\pi}{6}$ in degrees

ANSWER 150°

4. $\frac{7\pi}{4}$ in degrees

ANSWER 315°

5. $r = 6, \alpha = \frac{\pi}{3}$: arc

ANSWER 2π

6. $r = 6, \alpha = \frac{\pi}{3}$: sector area

ANSWER 6π

7. Arc 10, $\alpha = 2$: radius

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. 1 rad in degrees, to 1 d.p.

ANSWER 57.3°

2. Arc 12, radius 8: the angle in degrees

ANSWER $\approx 85.9^\circ$

3. A sector of area 18 and radius 6: the angle

ANSWER 1 rad

4. Perimeter of a sector, $r = 5$, $\alpha = \frac{\pi}{2}$

ANSWER $10 + \frac{5\pi}{2}$

5. A sector of angle $\frac{2\pi}{3}$ has area 12π : the radius

ANSWER 6

6. The minute hand is 9 cm; the arc its tip sweeps in 20 minutes

ANSWER 6π cm

7. Area swept by that hand in 20 minutes

ANSWER 27π cm²

07 Homework

Homework — 5 tasks

Give every radian answer as an exact multiple of π , not as a decimal.

- Convert 120° , 225° and 300° to radians.
- Convert $\frac{\pi}{5}$, $\frac{4\pi}{3}$ and $\frac{11\pi}{6}$ to degrees.
- A circle of radius 10 cm: find the arc and sector for $\alpha = \frac{\pi}{4}$.

4. An arc of 15 cm subtends 1.5 rad . Find the radius.
5. Find the perimeter of a sector with $r = 8$ and $\alpha = \frac{\pi}{3}$.

Rotation of a point about the origin

The angle leaves the triangle: a point turning on the unit circle can turn as far as it likes.

LESSON 41–42

2 × 40 MIN

ALGEBRA 9, §18

IGX 15.2

LESSON CLOCK

13'

20'

23'

20'

4'

An angle as a turn	13 min
Sign and size	20 min
Angles that land together	23 min
Quadrants	20 min
Homework	4 min

LEARNING OBJECTIVES

- Describe an angle as a rotation of the point P(1, 0) about the origin.
- Give a rotation a sign — anticlockwise positive, clockwise negative.
- Find every angle that lands a point in the same position: $\alpha + 2\pi k$.
- Name the quadrant in which a given rotation ends.

TEXTBOOK

National	pp. 97–102
Cambridge	Core & Extended, pp. 310–314
Workbook	Exercise 15.2

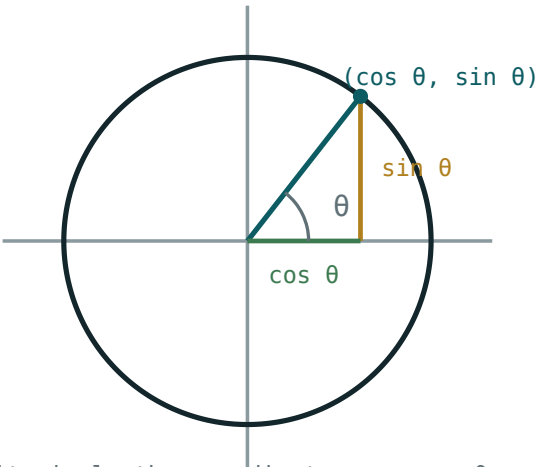
01

Explanation

An angle as a turn

In Grade 8 an angle lived inside a triangle, so it was between 0° and 180° . That restriction is now dropped.

Draw the **unit circle** $x^2 + y^2 = 1$ and start at $P_0(1, 0)$. Turn the point about the origin. The angle of the turn is the angle; the ray OP_0 is the **initial side** and the ray to the new position the **terminal side**.



on the unit circle the coordinates are $\cos \theta$ and $\sin \theta$

Every angle is a position reached by turning $P(1, 0)$ about the origin.

NOTHING IS LOST, AND A GREAT DEAL IS GAINED

For $0 < \alpha < \frac{\pi}{2}$ the new definition agrees exactly with the triangle one — the old sine and cosine are the coordinates of the turned point. But now $\alpha = 400^\circ$ and $\alpha = -30^\circ$ also mean something, and that is what makes trigonometry into a subject about functions rather than triangles.

Sign and size

TURN	SIGN	EXAMPLE	ENDS AT
anticlockwise	positive	$+90^\circ$	$(0, 1)$
clockwise	negative	-90°	$(0, -1)$
more than one turn	positive	450°	$(0, 1)$
no turn	zero	0	$(1, 0)$

The convention is universal and never varies: anticlockwise is the positive direction, in every country and every textbook. It matches the direction in which the first quadrant is entered.

A NEGATIVE ANGLE IS NOT A NEGATIVE NUMBER OF DEGREES “BACKWARDS ALONG THE AXIS”

-30° means turn thirty degrees the other way — clockwise, into the fourth quadrant. It is a direction, not a subtraction.

Angles that land together

A full turn is 2π , so adding or subtracting 2π changes the angle without moving the point.

α and $\alpha + 2\pi k, k \in \mathbb{Z}$, end at the same place

ANGLE	SAME POSITION AS	BECAUSE
400°	40°	$400 - 360$
-30°	330°	$-30 + 360$
$\frac{9\pi}{4}$	$\frac{\pi}{4}$	$\frac{9\pi}{4} - 2\pi$
$-\frac{\pi}{3}$	$\frac{5\pi}{3}$	$-\frac{\pi}{3} + 2\pi$
1080°	0°	three whole turns

THIS IS WHERE PERIODICITY COMES FROM

Because the position repeats every 2π , so does every quantity read off the position. That single observation is the reason sine and cosine are periodic with period 2π — a fact the next lesson will use constantly.

Quadrants

The two axes cut the circle into four quadrants, numbered anticlockwise from the top right.

QUADRANT	DEGREES	RADIANS	SIGNS OF (X, Y)
I	$0^\circ\text{--}90^\circ$	$0\text{--}\frac{\pi}{2}$	(+, +)
II	$90^\circ\text{--}180^\circ$	$\frac{\pi}{2}\text{--}\pi$	(-, +)
III	$180^\circ\text{--}270^\circ$	$\pi\text{--}\frac{3\pi}{2}$	(-, -)
IV	$270^\circ\text{--}360^\circ$	$\frac{3\pi}{2}\text{--}2\pi$	(+, -)

To find the quadrant of any angle, first reduce it to $[0, 2\pi)$ by adding or subtracting whole turns, then read the table.

THE AXES BELONG TO NO QUADRANT

90° , 180° , 270° and 0° are boundary angles. Asked for “the quadrant of 180° ”, the answer is that there is none — it lies on the negative x -axis.

02

Worked examples

EXAMPLE 1 Where does a rotation of 400° end, and in which quadrant?

① $400 - 360 = 40$
One full turn removed.

② So it ends where 40° ends.

③ $0^\circ < 40^\circ < 90^\circ$

④ Quadrant I.

ANSWER

Same as 40° — quadrant I

EXAMPLE 2 In which quadrant does $-\frac{2\pi}{3}$ end?

① Add a full turn: $-\frac{2\pi}{3} + 2\pi$

② $= \frac{4\pi}{3}$

③ $\pi < \frac{4\pi}{3} < \frac{3\pi}{2}$

④ Quadrant III.

ANSWER

Quadrant III

EXAMPLE 3 Write every angle that ends where 50° ends.

① Adding whole turns changes nothing.

② $50^\circ + 360^\circ k, k \in \mathbb{Z}$.

③ In radians: $\frac{5\pi}{18} + 2\pi k$.

④ $k = -1$ gives -310° — the same place.

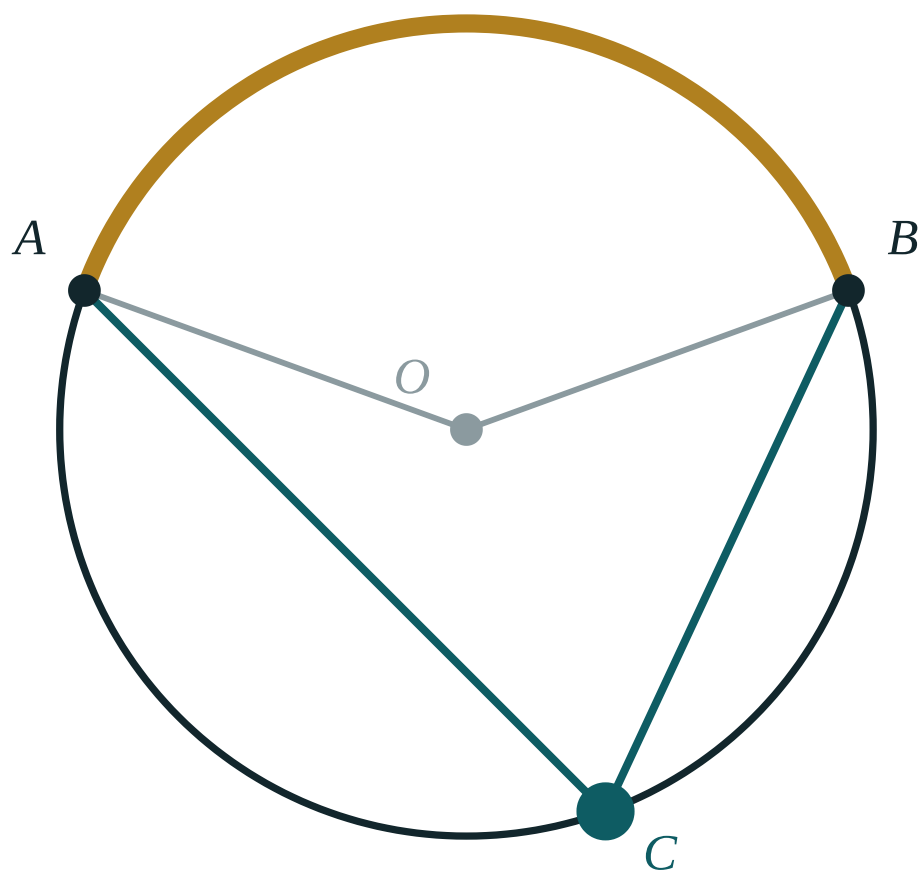
ANSWER

$50^\circ + 360^\circ k, k \in \mathbb{Z}$

03 Interactive model

Pin a paper arrow at the origin of a circle drawn on the board; turn it 400° and then -310° and let the class see both stop in the same place.

Turn the point



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rotation	Aylanish	Поворот
Unit circle	Birlik aylana	Единичная окружность
Initial side	Boshlang‘ich tomon	Начальная сторона
Terminal side	Yakuniy tomon	Конечная сторона
Anticlockwise	Soat strelkasiga teskari	Против часовой стрелки
Quadrant	Chorak	Четверть
Coterminal angles	Ustma-ust tushuvchi burchaklar	Совпадающие углы
Full turn	To‘liq aylanish	Полный оборот

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The positive direction of rotation is:

Rotation starts from the point:

400° ends where:

-30° ends where:

30°

150°

330°

210°

Angles ending together differ by:

πk

$2\pi k$

$\frac{\pi}{2} k$

πk^2

$\frac{4\pi}{3}$ lies in quadrant:

I

II

III

IV

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Quadrant of 50°

ANSWER I

2. Quadrant of 120°

ANSWER II

3. Quadrant of 200°

ANSWER III

4. Quadrant of 300°

ANSWER IV

5. 370° ends where?

ANSWER 10°

6. -90° ends where?

ANSWER 270°

7. Sign of an anticlockwise turn

ANSWER Positive

Medium · the standard question

7 PROBLEMS

1. Reduce 400° to $[0^\circ, 360^\circ)$

ANSWER 40°

2. Reduce -200° to $[0^\circ, 360^\circ)$

ANSWER 160°

3. Reduce $\frac{9\pi}{4}$ to $[0, 2\pi)$

ANSWER $\frac{\pi}{4}$

4. Reduce $-\frac{\pi}{3}$ to $[0, 2\pi)$

ANSWER $\frac{5\pi}{3}$

5. Quadrant of $\frac{4\pi}{3}$

ANSWER III

6. Quadrant of $-\frac{2\pi}{3}$

ANSWER III

7. All angles ending where 50° ends

ANSWER $50^\circ + 360^\circ k$

Hard · stretch and reason

7 PROBLEMS

1. Reduce 1000° to $[0^\circ, 360^\circ)$

ANSWER 280°

2. Reduce $-\frac{17\pi}{6}$ to $[0, 2\pi)$

ANSWER $\frac{7\pi}{6}$

3. Quadrant of -1000°

ANSWER I — it reduces to 80°

4. How many full turns in 2000° ?

ANSWER 5, with 200° left

5. Two angles end together and differ by 1440° . How many turns?

ANSWER 4

6. For which k does $30^\circ + 360^\circ k$ lie in $[700^\circ, 1100^\circ)$?

ANSWER $k = 2$ — that is 750°

7. A wheel turns 7.25 times. The final position, in degrees

ANSWER 90°

07 Homework

Homework — 5 tasks

For every angle give the reduced angle, the quadrant, and one negative angle ending in the same place.

- Reduce 520° , -140° and 900° to $[0^\circ, 360^\circ)$.
- Reduce $\frac{11\pi}{4}$ and $-\frac{5\pi}{3}$ to $[0, 2\pi)$.
- Name the quadrant of 160° , $\frac{7\pi}{6}$ and -80° .

4. Write every angle that ends where $\frac{\pi}{6}$ ends.
5. A wheel makes 3.75 turns. Where does the marked point finish?

Sine, cosine, tangent and cotangent of any angle

Four numbers read off one turned point — and the definitions no longer need a triangle.

LESSON 43–44

2 × 40 MIN

ALGEBRA 9, §19

IGX 15.3

LESSON CLOCK

16'

20'

20'

20'

4'

The definitions	16 min
The exact values	20 min
Where tangent and cotangent fail	20 min
The fundamental identity	20 min
Homework	4 min

LEARNING OBJECTIVES

- Define $\sin \alpha$ and $\cos \alpha$ as the coordinates of the point turned through α on the unit circle.
- Define $\tan \alpha$ and $\cot \alpha$ as the ratios y/x and x/y , and say where they fail to exist.
- Give the exact values at $0, \pi/6, \pi/4, \pi/3, \pi/2$ and the axis angles.
- State and use $\sin^2 \alpha + \cos^2 \alpha = 1$.

TEXTBOOK

National	pp. 103–108
Cambridge	Core & Extended, pp. 315–321
Workbook	Exercise 15.3

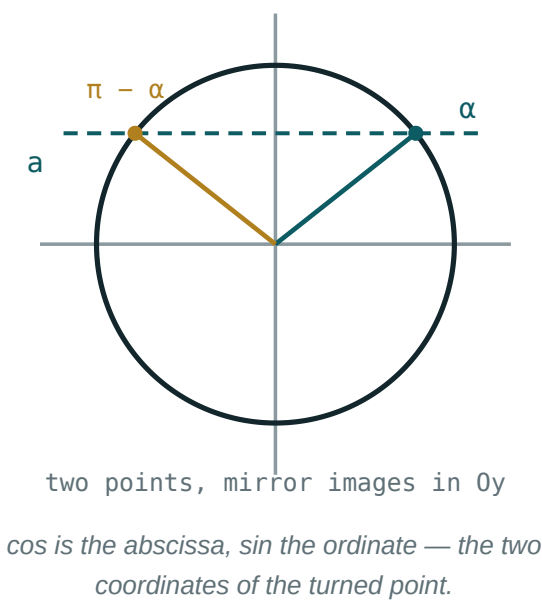
01

Explanation

The definitions

Turn $P_0(1, 0)$ through α and let it arrive at $P(x, y)$. The four quantities are read straight off that point.

$$\cos \alpha = x \quad \sin \alpha = y \quad \tan \alpha = \frac{y}{x} \quad \cot \alpha = \frac{x}{y}$$



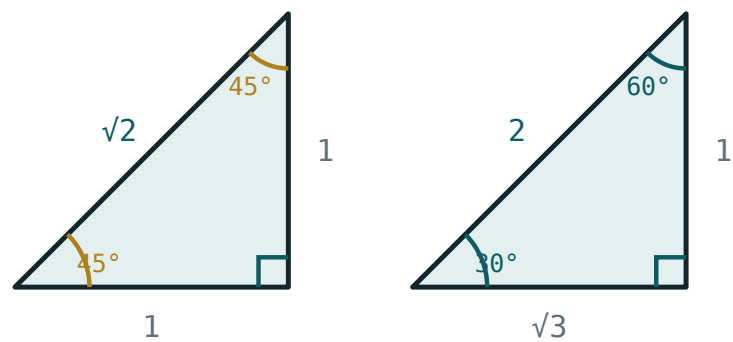
For an acute angle the point lies in the first quadrant and the right triangle OPQ has hypotenuse 1 , so $\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}} = y$ exactly as before. Nothing that was true in Grade 8 has changed; the definition has only been widened.

TWO COORDINATES, FOUR FUNCTIONS

Only $\sin$ and $\cos$ are genuinely new definitions. Tangent and cotangent are quotients of them: $\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$ and $\cot \alpha = \frac{\cos \alpha}{\sin \alpha}$. Every identity in the next chapter follows from that.

The exact values

A	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$
$\sin \alpha$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1
$\cos \alpha$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0
$\tan \alpha$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	$—$	0	$—$
$\cot \alpha$	$—$	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0	$—$	0



The half-square and the half-equilateral triangle — where every exact value comes from.

READ THE SINE ROW, NOT THE WHOLE TABLE

The sine row for $0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$ is $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$ — the roots of 0, 1, 2, 3, 4 over two. The cosine row is the same read backwards, and the tangent row is one divided by the other.

Where tangent and cotangent fail

A quotient needs a non-zero denominator, so two of the four functions have gaps.

FUNCTION	UNDEFINED WHEN	THAT IS	BECAUSE THE POINT IS AT
$\tan \alpha$	$\cos \alpha = 0$	$\alpha = \frac{\pi}{2} + \pi k$	$(0, \pm 1)$
$\cot \alpha$	$\sin \alpha = 0$	$\alpha = \pi k$	$(\pm 1, 0)$

Sine and cosine, by contrast, exist for every angle: they are coordinates, and a point on the circle always has both.

FUNCTION	DOMAIN	RANGE
$\sin \alpha, \cos \alpha$	$\mathbb{R}$	$[-1, 1]$
$\tan \alpha$	$\alpha \neq \frac{\pi}{2} + \pi k$	$\mathbb{R}$
$\cot \alpha$	$\alpha \neq \pi k$	$\mathbb{R}$

SIN A=2 IS NOT A HARD EQUATION – IT IS AN IMPOSSIBLE ONE

Both coordinates of a point on the unit circle lie between -1 and 1 . An equation asking for a sine or cosine outside that range has no solution at all, and saying so is the complete answer.

The fundamental identity

The point $P(x, y)$ lies on $x^2 + y^2 = 1$. Substituting the definitions gives the identity everything else is built on:

$\sin^2\alpha + \cos^2\alpha = 1$ for every angle α

It is Pythagoras' theorem, written for a triangle with hypotenuse 1 — and because it came from the circle rather than from a triangle, it holds for angles of any size and sign.

GIVEN	THEN	SIGN CHOSEN BY
$\sin \alpha = \frac{3}{5}$	$\cos \alpha = \pm \frac{4}{5}$	the quadrant
$\cos \alpha = -\frac{5}{13}$	$\sin \alpha = \pm \frac{12}{13}$	the quadrant

THE IDENTITY GIVES THE SIZE, THE QUADRANT GIVES THE SIGN

$\sin^2\alpha + \cos^2\alpha = 1$ always produces $\pm$. It cannot tell you which — only the quadrant can, and that is exactly what the next lesson is about. An answer left as $\pm$ when the quadrant was given is an incomplete answer.

02 Worked examples

EXAMPLE 1 Find $\sin$, $\cos$, $\tan$, $\cot$ of $\frac{\pi}{4}$.

① The point is $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$.

Half a square.

② $\cos = \frac{\sqrt{2}}{2}$, $\sin = \frac{\sqrt{2}}{2}$.

③ $\tan = \frac{y}{x} = 1$

④ $\cot = 1$
Equal coordinates.

ANSWER

$$\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1, 1$$

EXAMPLE 2 Given $\sin \alpha = \frac{3}{5}$ and α in quadrant II, find the other three.

$$\textcircled{1} \quad \cos^2 \alpha = 1 - \frac{9}{25} = \frac{16}{25}$$

$$\textcircled{2} \quad \text{Quadrant II: } \cos \alpha < 0, \text{ so } \cos \alpha = -\frac{4}{5}.$$

The sign comes from the quadrant.

$$\textcircled{3} \quad \tan \alpha = \frac{3/5}{-4/5} = -\frac{3}{4}$$

$$\textcircled{4} \quad \cot \alpha = -\frac{4}{3}$$

ANSWER

$$\cos = -\frac{4}{5}, \tan = -\frac{3}{4}, \cot = -\frac{4}{3}$$

EXAMPLE 3 Evaluate $\sin \frac{\pi}{2} + \cos \pi - \tan 0$.

$$\textcircled{1} \quad \sin \frac{\pi}{2} = 1$$

The point $(0, 1)$.

$$\textcircled{2} \quad \cos \pi = -1$$

The point $(-1, 0)$.

$$\textcircled{3} \quad \tan 0 = 0$$

$$\textcircled{4} \quad 1 + (-1) - 0 = 0$$

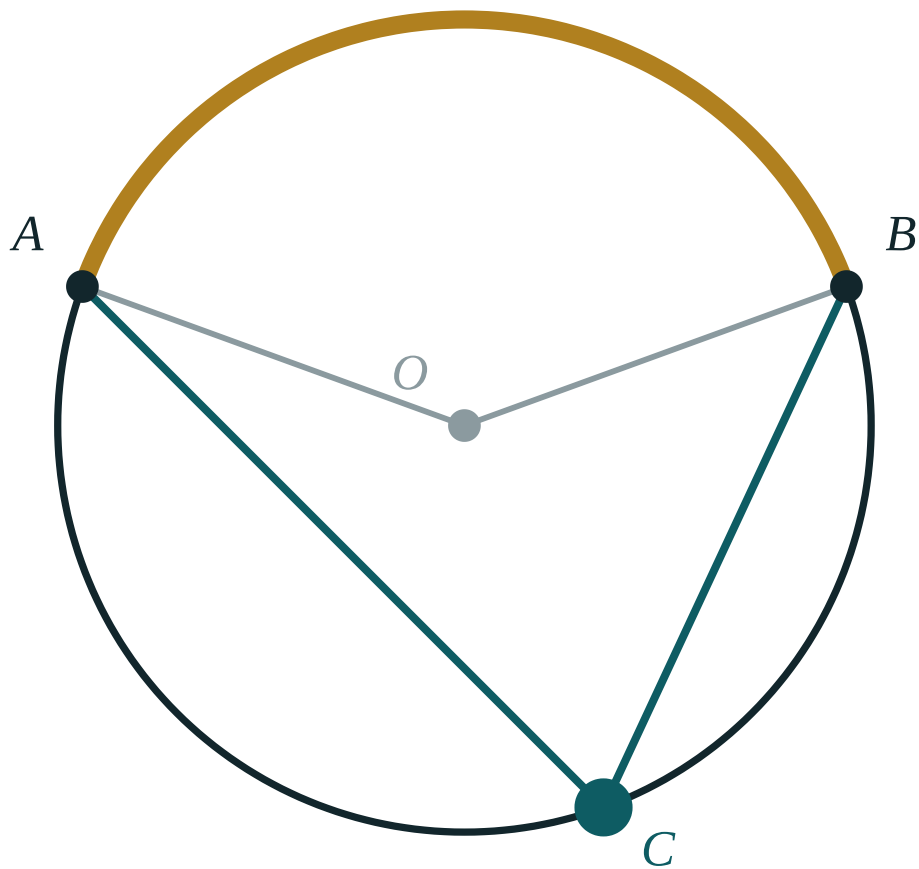
ANSWER

0

03 Interactive model

Draw the unit circle once and label only the point; ask the class to read each of the four values off it, so the definitions stay attached to a picture rather than to a table.

Read the four values off the point



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sine	Sinus	Синус
Cosine	Kosinus	Косинус
Tangent	Tangens	Тангенс
Cotangent	Kotangens	Котангенс
Ordinate	Ordinata	Ордината
Abscissa	Abssissa	Абсцисса
Undefined	Aniqlanmagan	Не определён
Fundamental identity	Asosiy ayniyat	Основное тождество

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\cos \alpha$ is the point's:

ordinate

abscissa

radius

arc

$\tan \alpha$ equals:

$$\frac{x}{y}$$

$$\frac{y}{x}$$

xy

$x + y$

$\sin \frac{\pi}{3}$ is:

$$\frac{1}{2}$$

$$\frac{\sqrt{2}}{2}$$

$$\frac{\sqrt{3}}{2}$$

1

$\tan \alpha$ is undefined when:

The range of $\cos \alpha$:

$\sin^2 \alpha + \cos^2 \alpha$ equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\sin 0$

ANSWER 0

2. $\cos 0$

ANSWER 1

3. $\sin \frac{\pi}{2}$

ANSWER 1

4. $\cos \pi$

ANSWER -1

5. $\tan \frac{\pi}{4}$

ANSWER 1

6. $\sin \frac{\pi}{6}$

ANSWER $\frac{1}{2}$

7. $\cos \frac{\pi}{3}$

ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $\cos \frac{\pi}{6}$

ANSWER $\frac{\sqrt{3}}{2}$

2. $\tan \frac{\pi}{3}$

ANSWER $\sqrt{3}$

3. $\cot \frac{\pi}{6}$

ANSWER $\sqrt{3}$

4. $\sin \frac{3\pi}{2}$

ANSWER -1

5. $\tan \frac{\pi}{2}$

ANSWER Undefined

6. $\sin^2 \frac{\pi}{4} + \cos^2 \frac{\pi}{4}$

ANSWER 1

7. $\sin \frac{\pi}{2} + \cos \pi - \tan 0$

ANSWER 0

Hard · stretch and reason

7 PROBLEMS

1. $\sin \alpha = \frac{3}{5}$, quadrant II: $\cos \alpha$

ANSWER $-\frac{4}{5}$

2. $\cos \alpha = -\frac{5}{13}$, quadrant III: $\sin \alpha$

ANSWER $-\frac{12}{13}$

3. $\tan \alpha = \frac{3}{4}$, quadrant III: $\sin \alpha$

ANSWER $-\frac{3}{5}$

4. Simplify $\sin^4 \alpha - \cos^4 \alpha$

ANSWER $\sin^2 \alpha - \cos^2 \alpha$

5. Simplify $(1 - \cos^2 \alpha)(1 + \cot^2 \alpha)$

ANSWER 1

6. Solve $\sin \alpha = 2$

ANSWER No solution

7. $\sin \alpha + \cos \alpha = \frac{1}{5}$: find $\sin \alpha \cos \alpha$

ANSWER $-\frac{12}{25}$

07 Homework

Homework — 5 tasks

Every exact value must be written as a fraction or a surd — no decimals in this homework.

1. Complete the table of $\sin$, $\cos$, $\tan$, $\cot$ for 0 , $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, $\frac{\pi}{2}$.

2. Given $\cos \alpha = \frac{8}{17}$ and α in quadrant IV, find the other three.
3. Evaluate $\cos \frac{\pi}{3} + \sin \frac{\pi}{6} - \tan \frac{\pi}{4}$.
4. Say why $\cos \alpha = 1.2$ has no solution.
5. Simplify $(1 - \sin^2 \alpha)(1 + \tan^2 \alpha)$.

The signs of the trigonometric functions in the quadrants

Which of the four are positive where — decided by two coordinates, not by a rhyme.

LESSON 45–46

2 × 40 MIN

ALGEBRA 9, §20

IGX 15.4

LESSON CLOCK

13'

20'

23'

20'

4'

Where the signs come from

13 min

The table

20 min

Choosing the sign

23 min

Working backwards

20 min

Homework

4 min

LEARNING OBJECTIVES

- Give the sign of each of the four functions in each of the four quadrants.
- Derive the sign table from the signs of x and y rather than memorising it.
- Choose the correct sign after using the fundamental identity.
- Find the quadrant of an angle from the signs of two of its functions.

TEXTBOOK

National

pp. 109–113

Cambridge

Core & Extended, pp. 322–327

Workbook

Exercise 15.4

01

Explanation

Where the signs come from

There is nothing to learn here that is not already known: $\cos \alpha = x$ and $\sin \alpha = y$, so their signs are simply the signs of the coordinates.

QUADRANT	X	Y	$\cos$	$\sin$
I	+	+	+	+
II	–	+	–	+
III	–	–	–	–
IV	+	–	+	–

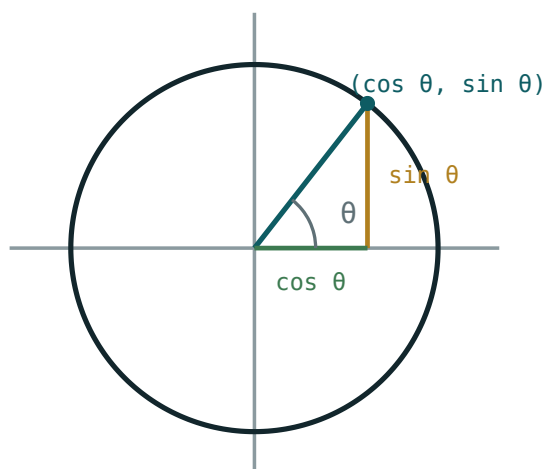
Tangent and cotangent are quotients of the two, so they are positive exactly where the coordinates share a sign — quadrants I and III.

DO NOT MEMORISE A MNEMONIC YOU CANNOT REBUILD

Rhymes for this table exist in every language and they all fail under pressure, because a rhyme cannot be checked. Sketching the circle and asking “is x left or right of the axis?” takes four seconds and is never wrong.

The table

FUNCTION	I	II	III	IV	POSITIVE IN
$\sin \alpha$	+	+	−	−	the upper half
$\cos \alpha$	+	−	−	+	the right half
$\tan \alpha$	+	−	+	−	I and III
$\cot \alpha$	+	−	+	−	I and III



on the unit circle the coordinates are $\cos \theta$ and $\sin \theta$

*sine is positive above the axis, cosine to the right of it
— the two halves that decide everything.*

Two sentences replace the whole table: **sine is positive above the x -axis; cosine is positive to the right of the y -axis.** Tangent and cotangent follow by division.

Choosing the sign

This is the working use of the table. The identity $\sin^2 \alpha + \cos^2 \alpha = 1$ always leaves a $\pm$; the quadrant removes it.

Example. $\cos \alpha = -\frac{5}{13}$ and α is in quadrant III. Then $\sin^2 \alpha = 1 - \frac{25}{169} = \frac{144}{169}$, so $\sin \alpha = \pm \frac{12}{13}$ — and quadrant III makes it negative: $\sin \alpha = -\frac{12}{13}$.

STEP	GIVES
the identity	the size
the quadrant	the sign
the quotient	$\tan, \cot$

TWO ANSWERS WHERE THE QUESTION EXPECTED ONE

If the quadrant is **not** given, $\pm$ is the correct answer and both cases must be written out. If it **is** given and you still write $\pm$, the mark is lost — the whole point of the quadrant was to decide.

Working backwards

Two signs pin down the quadrant, because each function splits the circle into two halves and two halves intersect in one quarter.

GIVEN	QUADRANT	WHY
$\sin \alpha > 0, \cos \alpha < 0$	II	upper half, left half
$\sin \alpha < 0, \cos \alpha > 0$	IV	lower half, right half
$\tan \alpha > 0, \sin \alpha < 0$	III	matching signs, lower half
$\cos \alpha > 0, \tan \alpha < 0$	I	right half, mismatched signs

One sign alone is never enough: $\sin \alpha > 0$ leaves quadrants I and II both open.

ON AN AXIS THERE IS NO QUADRANT

At $\alpha = \pi$ the point is $(-1, 0)$: $\sin \alpha = 0$, which is neither positive nor negative, and $\cot \alpha$ does not exist. Boundary angles are handled by reading the point, not by the sign table.

02 Worked examples

EXAMPLE 1 Give the sign of $\sin 200^\circ$, $\cos 200^\circ$ and $\tan 200^\circ$.

① $180^\circ < 200^\circ < 270^\circ$ — quadrant III.

② Both coordinates negative.

③ $\sin < 0$, $\cos < 0$.

④ $\tan > 0$ — matching signs.

ANSWER

$-$, $-$, $+$

EXAMPLE 2 Given $\cos \alpha = -\frac{5}{13}$ with α in quadrant III, find $\sin \alpha$ and $\tan \alpha$.

$$\textcircled{1} \quad \sin^2 \alpha = 1 - \frac{25}{169} = \frac{144}{169}$$

$$\textcircled{2} \quad \sin \alpha = \pm \frac{12}{13}$$

The size.

$$\textcircled{3} \quad \text{Quadrant III} \Rightarrow \text{negative: } \sin \alpha = -\frac{12}{13}.$$

The sign.

$$\textcircled{4} \quad \tan \alpha = \frac{-12/13}{-5/13} = \frac{12}{5}$$

Positive, as III demands.

ANSWER

$$\sin \alpha = -\frac{12}{13}, \tan \alpha = \frac{12}{5}$$

EXAMPLE 3 In which quadrant is α if $\tan \alpha > 0$ and $\cos \alpha < 0$?

$$\textcircled{1} \quad \tan \alpha > 0 \Rightarrow \text{quadrant I or III.}$$

Matching signs.

$$\textcircled{2} \quad \cos \alpha < 0 \Rightarrow \text{quadrant II or III.}$$

Left half.

$$\textcircled{3} \quad \text{The only quadrant in both lists is III.}$$

$$\textcircled{4} \quad \text{Quadrant III.}$$

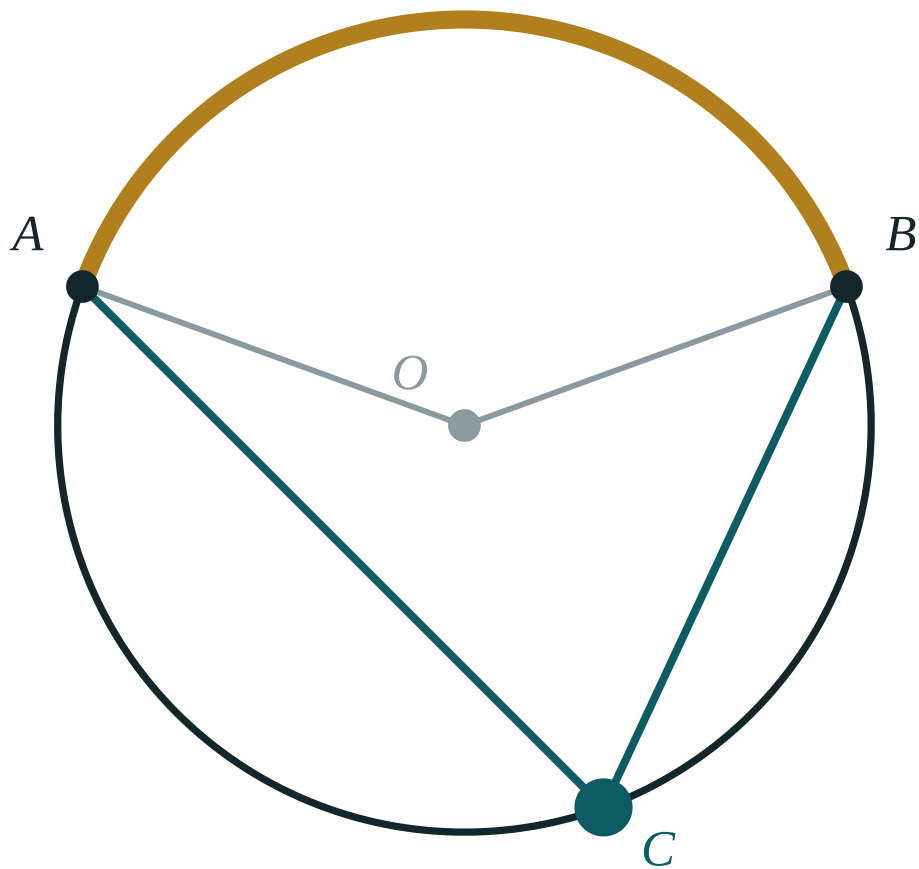
ANSWER

Quadrant III

03 Interactive model

Ask for the sign of $\sin 200^\circ$ before any algebra: the class should point at the lower-left of a sketched circle, not recite a rule.

Signs around the circle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sign	Ishora	Знак
Quadrant	Chorak	Четверть
Positive	Musbat	Положительный
Negative	Manfiy	Отрицательный
Coordinate	Koordinata	Координата
Quotient	Bo'linma	Частное
Boundary angle	Chegaraviy burchak	Граничный угол
Determine	Aniqlash	Определить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin \alpha$ is positive in quadrants:

I and II

I and III

III and IV

II and III

$\cos \alpha$ is positive in quadrants:

I and II

I and IV

II and III

III and IV

$\tan \alpha$ is positive in quadrants:

I and II

I and III

II and IV

III and IV

Sign of $\cos 200^\circ$:

positive

negative

zero

undefined

$\sin \alpha > 0$ and $\cos \alpha < 0$ means quadrant:

I

II

III

IV

The identity gives the size; the sign comes from:

the identity

the quadrant

the calculator

the radius

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Sign of $\sin 100^\circ$

ANSWER +

2. Sign of $\cos 100^\circ$

ANSWER -

3. Sign of $\sin 250^\circ$

ANSWER -

4. Sign of $\cos 300^\circ$

ANSWER +

5. Sign of $\tan 200^\circ$

ANSWER +

6. Sign of $\tan 150^\circ$

ANSWER -

7. Quadrant of α if both are positive

ANSWER I

Medium · the standard question

7 PROBLEMS

1. Signs of $\sin$, $\cos$, $\tan$ at 200°

ANSWER $-, -, +$

2. Signs at $\frac{3\pi}{4}$

ANSWER $+, -, -$

3. Signs at $-\frac{\pi}{4}$

ANSWER $-, +, -$

4. $\sin \alpha > 0$, $\tan \alpha < 0$: quadrant

ANSWER II

5. $\cos \alpha > 0$, $\sin \alpha < 0$: quadrant

ANSWER IV

6. $\cos \alpha = -\frac{5}{13}$, III: $\sin \alpha$

ANSWER $-\frac{12}{13}$

7. $\sin \alpha = \frac{4}{5}$, II: $\tan \alpha$

ANSWER $-\frac{4}{3}$

Hard · stretch and reason

7 PROBLEMS

1. $\tan \alpha = \frac{3}{4}$, III: $\sin \alpha$ and $\cos \alpha$

ANSWER $-\frac{3}{5}, -\frac{4}{5}$

2. $\cos \alpha = 0.6$, IV: $\cot \alpha$

ANSWER $-\frac{3}{4}$

3. $\sin \alpha \cos \alpha < 0$: which quadrants?

ANSWER II and IV

4. $\sin \alpha + \cos \alpha$ at $\frac{5\pi}{4}$

ANSWER $-\sqrt{2}$

5. Sign of $\sin 400^\circ \cdot \cos 400^\circ$

ANSWER $+$ — quadrant I

6. $\sin \alpha = -\frac{7}{25}$, IV: $\cos \alpha$

ANSWER $\frac{24}{25}$

7. For which quadrants is $\sin \alpha + \cos \alpha$ certainly negative?

ANSWER III only

07 Homework

Homework — 5 tasks

Sketch the circle for every task; a sign written without a picture is a guess.

1. Give the signs of the four functions at 130° , 220° and 310° .

2. Given $\sin \alpha = -\frac{8}{17}$ with α in quadrant III, find $\cos \alpha$ and $\tan \alpha$.

3. In which quadrant is α if $\cot \alpha > 0$ and $\sin \alpha < 0$?
4. Explain why one sign alone never determines the quadrant.
5. Find $\sin \alpha + \cos \alpha$ at $\frac{7\pi}{4}$.

Control work 4, and the quarter reviewed

The radian, the turned point and the four functions — the opening of trigonometry, tested.

LESSON 47–48

2 × 40 MIN

ALGEBRA 9, NAZORAT ISHI 4

IGX 15 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The quarter	5 min

LEARNING OBJECTIVES

- Convert between degrees and radians and use $l = r\alpha$, $S = \frac{1}{2}r^2\alpha$ under time.
- Reduce an angle to $[0, 2\pi)$ and name its quadrant.
- Use the fundamental identity together with the quadrant to fix a sign.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 91–113
Cambridge	Core & Extended, pp. 304–330
Workbook	Control paper A4

01

Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Convert 150° to radians and $\frac{7\pi}{6}$ to degrees	4	L39–40
2	$r = 9$, $\alpha = \frac{\pi}{3}$: find the arc and the sector area	6	L39–40
3	Reduce -480° to $[0^\circ, 360^\circ)$ and name the quadrant	5	L41–42
4	Evaluate $\sin \frac{\pi}{6} + \cos \frac{\pi}{3} - \tan \frac{\pi}{4}$	5	L43–44
5	$\sin \alpha = \frac{12}{13}$, quadrant II: find $\cos \alpha$ and $\tan \alpha$	6	L45–46
6	In which quadrant is α if $\tan \alpha < 0$ and $\sin \alpha > 0$?	4	L45–46

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for cancelling the fraction; Q2 one for using radians rather than $\frac{\theta}{360}$; Q3 one for adding **two** full turns; Q5 two marks for the sign, not the size; Q6 one for naming both halves before intersecting them.

Naming the slip

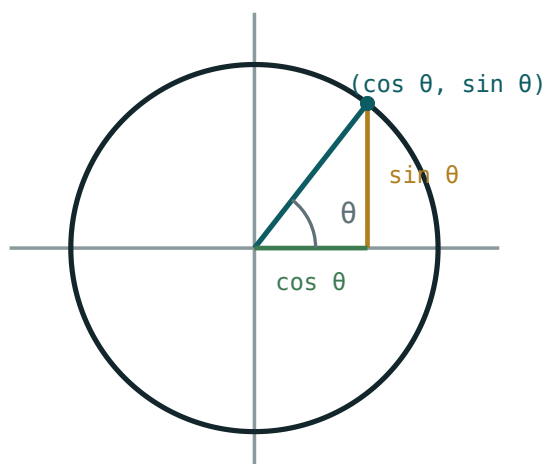
SLIP	WHAT IT LOOKS LIKE	THE FIX
fraction not cancelled	$\frac{150\pi}{180}$	$\frac{5\pi}{6}$
degrees in $l = r\alpha$	$l = 9 \cdot 60$	convert first
one turn only	$-480 + 360 = -120$	add 720°
exact value guessed	$\sin \frac{\pi}{6} = \frac{\sqrt{3}}{2}$	$\frac{1}{2}$ — <i>sine of the smaller angle is smaller</i>
sign left as $\pm$	$\cos \alpha = \pm \frac{5}{13}$	quadrant II $\Rightarrow$ negative
one condition used	$\tan \alpha < 0 \Rightarrow \text{II}$	II or IV; the second condition decides
calculator in the wrong mode	$\sin \frac{\pi}{6} = 0.0091$	set RAD

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The quarter in one page

Quarter II covered two chapters, and each reduces to a single sentence.

BLOCK	THE SENTENCE
systems of equations	substitute, add, or use $u = x + y$, $v = xy$ — the shape chooses
systems of inequalities	solve separately, draw one number line, intersect
proving inequalities	everything to one side, then a square
the radian	angle = arc $\div$ radius, and $180^\circ = \pi$
rotation	an angle is a turn of $P(1, 0)$; $+2\pi k$ changes nothing
the four functions	$\cos = x$, $\sin = y$, and the other two are quotients
the signs	sine above the axis, cosine to the right of it



on the unit circle the coordinates are $\cos \theta$ and $\sin \theta$

The picture that carries the whole of the second half of this quarter.

LOOKING FORWARD

Quarter III takes the four functions and finds the relations between them: the identities, what happens to $-\alpha$, the addition formulae, the double angle, and the reduction formulae. Every one of them is proved on the circle drawn above, so the picture is worth keeping in the front of the exercise book.

02 Worked examples

EXAMPLE 1 Model answer, Q2: $r = 9$, $\alpha = \frac{\pi}{3}$.

$$① \quad l = r\alpha = 9 \cdot \frac{\pi}{3}$$

The angle is already in radians.

$$② \quad l = 3\pi \approx 9.42$$

$$③ \quad S = \frac{1}{2} \cdot 81 \cdot \frac{\pi}{3}$$

$$④ \quad S = \frac{27\pi}{2} \approx 42.41$$

ANSWER

$$l = 3\pi, S = \frac{27\pi}{2}$$

EXAMPLE 2 Model answer, Q3: reduce -480° and name the quadrant.

$$① \quad -480 + 360 = -120 \text{ — still negative.}$$

One turn is not enough.

$$② \quad -120 + 360 = 240$$

A second turn.

$$③ \quad 180^\circ < 240^\circ < 270^\circ$$

$$④ \quad \text{Quadrant III.}$$

ANSWER

240° — quadrant III

EXAMPLE 3 Model answer, Q5: $\sin \alpha = \frac{12}{13}$, quadrant II.

$$\textcircled{1} \quad \cos^2 \alpha = 1 - \frac{144}{169} = \frac{25}{169}$$

$$\textcircled{2} \quad \cos \alpha = \pm \frac{5}{13}$$

The size.

$$\textcircled{3} \quad \text{Quadrant II} \Rightarrow \cos \alpha = -\frac{5}{13}.$$

The sign.

$$\textcircled{4} \quad \tan \alpha = -\frac{12}{5}$$

Negative, as II demands.

ANSWER

$$\cos \alpha = -\frac{5}{13}, \tan \alpha = -\frac{12}{5}$$

03 Interactive model

Put Q5 on the board with the answer left as $\pm$, and ask the class what is still missing — the sign, and the sentence that justifies it.

Quarter II in twelve questions

A symmetric system suggests:

$$u = x + y, v = xy$$

squaring

a graph

addition

A candidate pair is tested in:

one equation

both

neither

the simpler one

A system of inequalities gives:

a union

an intersection

one interval always

the empty set

A denominator zero is drawn:

filled

open

either

twice

To prove $A \geq B$:

test values

show $A - B \geq 0$

draw it

show $A > 0$

Equality in $a + b \geq 2\sqrt{ab}$:

$a = 0$

$a = b$

$b = 1$

never

180° in radians:

$$\frac{\pi}{2}$$

$$\pi$$

$$2\pi$$

$$90$$

Arc length in radians:

$$l = r\alpha$$

$$l = r^2\alpha$$

$$l = 2\pi r\alpha$$

$$l = \frac{r}{\alpha}$$

400° ends where:

$$40^\circ$$

$$60^\circ$$

$$-40^\circ$$

$$340^\circ$$

$\cos \alpha$ is the:

ordinate

abscissa

radius

arc

$\tan \alpha$ is undefined when:

$\sin \alpha = 0$

$\cos \alpha = 0$

$\alpha = 0$

never

$\sin \alpha > 0$, $\cos \alpha < 0$ means quadrant:

I

II

III

IV

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Radian	Radian	Радиян
Sector	Sektor	Сектор
Rotation	Aylanish	Поворот
Quadrant	Chorak	Четверть
Fundamental identity	Asosiy ayniyat	Основное тождество
Exact value	Aniq qiymat	Точное значение
Revision	Takrorlash	Повторение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 loses a mark most often for:

a wrong formula

an uncanceled fraction

no units

a decimal

$l = r\alpha$ requires α in:

degrees

radians

either

turns

-480° needs:

one turn

two turns

three turns

none

In Q5 the quadrant supplies:

the size

the sign

both

neither

One sign condition determines:

one quadrant

two quadrants

all four

none

Work on the mistakes means:

fix the wrong line

rewrite the solution

copy the answer

skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 150° in radiansANSWER $\frac{5\pi}{6}$ 2. $\frac{7\pi}{6}$ in degreesANSWER 210° 3. $r = 9, \alpha = \frac{\pi}{3}$: arcANSWER 3π 4. $r = 9, \alpha = \frac{\pi}{3}$: sector areaANSWER $\frac{27\pi}{2}$ 5. Reduce -480° ANSWER 240° 6. $\sin \frac{\pi}{6} + \cos \frac{\pi}{3}$ ANSWER 1 7. Sign of $\cos 130^\circ$ ANSWER $-$

Medium · the standard question

7 PROBLEMS

1. $\sin \frac{\pi}{6} + \cos \frac{\pi}{3} - \tan \frac{\pi}{4}$

ANSWER 0

2. $\sin \alpha = \frac{12}{13}$, II: $\cos \alpha$

ANSWER $-\frac{5}{13}$

3. $\sin \alpha = \frac{12}{13}$, II: $\tan \alpha$

ANSWER $-\frac{12}{5}$

4. $\tan \alpha < 0$, $\sin \alpha > 0$: quadrant

ANSWER II

5. Quadrant of -480°

ANSWER III

6. Solve $\{ x + y = 5 ; xy = 6 \}$

ANSWER $(2, 3), (3, 2)$

7. Solve $\{ x^2 - 1 > 0 ; x^2 - 9 < 0 \}$

ANSWER $(-3, -1) \cup (1, 3)$

Hard · stretch and reason

7 PROBLEMS

1. A sector of radius 12 has area 24π : the angle

ANSWER $\frac{\pi}{3}$

2. Perimeter of that sector

ANSWER $24 + 4\pi$

3. $\cos \alpha = -\frac{3}{5}$, III: $\sin \alpha + \cos \alpha$

ANSWER $-\frac{7}{5}$

4. Prove $x^2 + 16 \geq 8x$ and name the equality case

ANSWER $(x - 4)^2 \geq 0$; $x = 4$

5. Solve $\{x^2 + y^2 = 25 ; x^2 - y^2 = 7\}$

ANSWER $(\pm 4, \pm 3)$

6. Reduce $-\frac{19\pi}{6}$ to $[0, 2\pi)$

ANSWER $\frac{5\pi}{6}$

7. A wheel of radius 30 cm rolls 6 m. The angle turned, in radians

ANSWER 20

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark; the holiday is not the place to leave them.

- Convert 240° to radians and $\frac{5\pi}{4}$ to degrees.
- A sector has $r = 12$ and $\alpha = \frac{\pi}{4}$. Find its arc, area and perimeter.

3. Reduce -750° to $[0^\circ, 360^\circ)$ and name the quadrant.
4. Given $\cos \alpha = -\frac{4}{5}$ with α in quadrant II, find $\sin \alpha$ and $\cot \alpha$.
5. Evaluate $\sin \frac{\pi}{3} \cdot \cos \frac{\pi}{6} + \cos \frac{\pi}{3} \cdot \sin \frac{\pi}{6}$.